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Journal of the American Statistical Association

ISSN: 0162-1459 (Print) 1537-274X (Online) Journal homepage: www.tandfonline.com/journals/uasa20

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To cite this article: Ye Tian, Haolei Weng, Lucy Xia & Yang Feng (29 Jun 2026): Robust Unsupervised Multi-Task and Transfer Learning on Gaussian Mixture Models, Journal of the American Statistical Association, DOI: 10.1080/01621459.2026.2670031

To link to this article: https://doi.org/10.1080/01621459.2026.2670031

JOURNAL OF THE AMERICAN STATISTICAL ASSOCIATION
2026, VOL. 00, NO. 0, 1–14: Theory and Methods
https://doi.org/10.1080/01621459.2026.2670031

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Robust Unsupervised Multi-Task and Transfer Learning on Gaussian Mixture Models

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ABSTRACT
Unsupervised learning has been widely used in many real-world applications. One of the simplest and most important unsupervised learning models is the Gaussian mixture model (GMM). In this work, we study the multi-task learning problem on GMMs, which aims to leverage potentially similar GMM parameter structures among tasks to obtain improved learning performance compared to single-task learning. We propose a multi-task GMM learning procedure based on the EM algorithm that effectively uses unknown similarities between related tasks and is robust against a fraction of outlier tasks from arbitrary distributions. The proposed procedure is shown to achieve the minimax optimal rate of convergence for both the parameter estimation error and the excess mis-clustering error, in a wide range of regimes. Moreover, we generalize our approach to tackle the problem of transfer learning for GMMs, where similar theoretical results are derived. Additionally, iterative unsupervised multi-task and transfer learning methods may suffer from an initialization alignment problem, and two alignment algorithms are proposed to resolve the issue. Finally, we demonstrate the effectiveness of our methods through simulations and real data examples. To the best of our knowledge, this is the first work studying multi-task and transfer learning on GMMs with theoretical guarantees. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

ARTICLE HISTORY
Received December 2024
Accepted April 2026

KEYWORDS
EM algorithm; Gaussian mixture models; Minimax rate; Multi-task learning; Robustness; Transfer learning; Unsupervised learning

1. Introduction

1.1. Gaussian Mixture Models (GMMs)

Unsupervised learning that learns patterns from unlabeled data is a prevalent problem in statistics and machine learning. Clustering is one of the most important problems in unsupervised learning, where the goal is to group the observations based on some similarity metrics. Researchers have developed numerous clustering methods including k-means (Forgy 1965), k-medians (Jain and Dubes 1988), spectral clustering (Ng, Jordan, and Weiss 2001), and hierarchical clustering (Murtagh and Contreras 2012), among others. On the other hand, clustering problems have been analyzed from the perspective of the mixture of several probability distributions (Scott and Symons 1971). The mixture of Gaussian distributions is one of the simplest models in this category and has been widely applied in many real applications (Yang and Ahuja 1998; Lee et al. 2012).

In the binary Gaussian mixture models (GMMs) with common covariances, each observation $Z \in \mathbb{R}^p$ comes from the following mixture of two Gaussian distributions:

$$Y = \begin{cases} 1, & \text{with probability } 1-w, \\ 2, & \text{with probability } w, \end{cases}$$

$$Z|Y = r \sim \mathcal{N}(\boldsymbol{\mu}_r, \boldsymbol{\Sigma}), \quad r = 1, 2,$$

where $w \in (0,1)$, $\boldsymbol{\mu}_1 \in \mathbb{R}^p$, $\boldsymbol{\mu}_2 \in \mathbb{R}^p$ and $\boldsymbol{\Sigma} \succ 0$ are parameters. This is the same setting as the linear discriminant analysis (LDA) problem in classification (Hastie, Tibshirani, and Friedman 2009), except that the label Y is unknown in the clustering problem, while it is observed in the classification case. It has been shown that the Bayes classifier for the LDA problem is

$$\mathcal{C}(\boldsymbol{z}) = \begin{cases} 1, & \text{if } \boldsymbol{\beta}^\top \boldsymbol{z} - \delta \le \log\left(\frac{1-w}{w}\right); \\ 2, & \text{otherwise}, \end{cases} \tag{1}$$

where $\boldsymbol{\beta} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) \in \mathbb{R}^p$ and $\delta = \boldsymbol{\beta}^\top(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)/2$. Note that $\boldsymbol{\beta}$ is usually referred to as the discriminant coefficient (Anderson 1958; Efron 1975). Naturally, this classifier is useful in clustering too. In clustering, after learning w, $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$, and $\boldsymbol{\beta}$, we can plug their estimators into (1) to group any new observation Z^{new}. Generally, we define the mis-clustering error rate of any given clustering method $\mathcal{C}$ as

$$R(\mathcal{C}) = \min_{\pi:\{1,2\}\to\{1,2\}} \mathbb{P}(\mathcal{C}(Z^{\text{new}}) \neq \pi(Y^{\text{new}})),$$

where π is a permutation function, Y^{new} is the label of a future observation Z^{new}, and the probability is taken w.r.t. the joint distribution of $(Z^{\text{new}}, Y^{\text{new}})$ based on parameters w, $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$ and $\boldsymbol{\Sigma}$. Here the error is calculated up to a permutation due to the lack of label information. It is clear that in the ideal case where the parameters are known, $\mathcal{C}(\cdot)$ in (1) achieves the optimal mis-clustering error. Multi-cluster Gaussian mixture models with $R \ge 3$ components can be described in a similar way. To maintain simplicity and highlight key intuitions, we focus on binary

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Supplementary materials for this article are available online. Please go to www.tandfonline.com/r/JASA.

GMMs in the main text, while a detailed analysis of the multi-cluster scenario is presented in Section S.2 of the supplementary materials.

There is a large volume of published studies on learning a GMM. The vast majority of approaches can be roughly divided into three categories. The first category is the method of moments, where the parameters are estimated through several moment equations (Pearson 1894; Kalai, Moitra, and Valiant 2010; Hsu and Kakade 2013; Ge, Huang, and Kakade 2015). The second category is the spectral method, where the estimation is based on the spectral decomposition (Vempala and Wang 2004; Hsu and Kakade 2013; Jin, Ke, and Wang 2017). The last category is the likelihood-based method, which includes the popular expectation-maximization (EM) algorithm as a canonical example. The general form of the EM algorithm was formalized by Dempster et al. (1977) in the context of incomplete data, though earlier works (Hartley 1958; Hasselblad 1966; Baum et al. 1970; Sundberg 1974) have studied EM-style algorithms in various concrete settings. Classical convergence results on the EM algorithm (Wu 1983; Redner and Walker 1984; Meng and Rubin 1994; McLachlan and Krishnan 2007) guarantee local convergence of the algorithm to fixed points of the sample likelihood. Recent advances in the analysis of the EM algorithm and its variants provide stronger guarantees by establishing geometric convergence rates of the algorithm to the underlying true parameters under mild initialization conditions. See, for example, Dasgupta and Schulman (2000), Wang et al. (2014), Xu, Hsu, and Maleki (2016), Balakrishnan, Wainwright, and Yu (2017), Yan, Yin, and Sarkar (2017), Cai, Ma, and Zhang (2019), Kwon and Caramanis (2020), and Zhao, Li, and Sun (2020) for GMM-related works. In this article, we propose modified versions of the EM algorithm with similarly strong guarantees to learn GMMs, under the new multi-task and transfer learning context.

1.2. Multi-Task Learning and Transfer Learning

Humans naturally transfer knowledge across related tasks, often learning them more efficiently together than in isolation. Multi-task learning (MTL) embodies this principle as a machine learning paradigm: multiple related tasks are learned jointly, exploiting their shared structure to improve generalization and sample efficiency relative to learning each task independently (Zhang and Yang 2021). There has been much research on MTL, which can be classified into five categories (Zhang and Yang 2021): feature learning approach (Obozinski, Taskar, and Jordan 2006; Argyriou, Evgeniou, and Pontil 2008), low-rank approach (Ando, Zhang, and Bartlett 2005), task clustering approach (Thrun and O'Sullivan 1996), task relation learning approach (Evgeniou and Pontil 2004) and decomposition approach (Jalali et al. 2010). Most existing works focus on using MTL in supervised learning problems, while the application of MTL in unsupervised learning, such as clustering, has received less attention. Zhang and Zhang (2011) developed an MTL clustering method based on a penalization framework, where the objective function consists of a local loss function and a pairwise task regularization term, both of which are related to the Bregman divergence. In Gu, Li, and Han (2011), a reproducing kernel Hilbert space (RKHS) was first established, and then a multi-task kernel k-means clustering was applied based on that RKHS. Yang et al. (2014) proposed a spectral MTL clustering method with a novel $\ell_{2,p}$-norm, which can also produce a linear regression function to predict labels for out-of-sample data. Zhang et al. (2018) suggested a new method based on the similarity matrix of samples in each task, which can learn the within-task clustering structure as well as the task relatedness simultaneously. Marfoq et al. (2021) established a new federated multi-task EM algorithm to learn the mixture of distributions and provided some theory on the convergence guarantee, but the statistical properties of the estimators were not fully understood. Zhang and Chen (2022) proposed a distributed learning algorithm for GMMs based on transportation divergence when all GMMs are identical. In general, there are very few theoretical results about unsupervised MTL.

Transfer learning (TL) is another learning paradigm similar to multi-task learning but has different objectives. While MTL aims to learn all the tasks well with no priority for any specific task, the goal of TL is to improve the performance on the *target* task using the information from the *source* tasks (Zhang and Yang 2021). According to Pan and Yang (2009), most TL approaches can be classified into four categories: instance-based transfer (Dai et al. 2007), feature representation transfer (Dai et al. 2008), parameter transfer (Lawrence and Platt 2004) and relational-knowledge transfer (Mihalkova, Huynh, and Mooney 2007). Similar to MTL, most of the TL methods focus on supervised learning. Some TL approaches are also developed for the semi-supervised learning setting (Chattopadhyay et al. 2012; Li et al. 2013), where only part of the target or source data is labeled. There is much less discussion on the unsupervised TL approaches,[1] which focus on the cases where both target and source data are unlabeled. Dai et al. (2008) developed a co-clustering approach to transfer information from a single source to the target, which relies on the loss of mutual information and requires the features to be discrete. Wang, Song, and Zhang (2008) proposed a TL discriminant analysis method, where the target data is allowed to be unlabeled, but some labeled source data is necessary. In Wang et al. (2021), a TL approach was developed to learn Gaussian mixture models with only one source by weighting the target and source likelihood functions. Zuo et al. (2018) proposed a TL method based on infinite Gaussian mixture models and active learning, but their approach needs sufficient labeled source data and a few labeled target samples.

There are some recent studies on TL and MTL under various statistical settings, including high-dimensional linear regression (Xu and Bastani 2021; Zhang et al. 2022; Li, Cai, and Li 2022a, 2022b), high-dimensional generalized linear models (Bastani 2021; Li, Cai, and Duan 2023; Tian and Feng 2023), functional linear regression (Lin and Reimherr 2022), high-dimensional graphical models (Li, Cai, and Li 2022b), reinforcement learning (Chen, Jordan, and Li 2022), among others. The recent work Duan and Wang (2023) developed an adaptive and robust MTL framework with sharp statistical guarantees for a broad class of models. We discuss its connection to our work in Section 2.

[1]There are different definitions for unsupervised TL. Sometimes, people call the semi-supervised TL an unsupervised TL as well. We follow the definition in Pan and Yang (2009) here.

1.3. Our Contributions and Article Structure

Our main contributions in this work can be summarized as follows:

(i) We develop efficient polynomial-time iterative procedures to learn GMMs in both MTL and TL settings. These procedures can be viewed as adaptations of the standard EM algorithm for MTL and TL problems.
(ii) The developed procedures come with provable statistical guarantees. Specifically, we derive the upper bounds of their estimation and excess mis-clustering error rates under mild conditions. For MTL, it is shown that when the tasks are close to each other, our method can achieve better upper bounds than those from the single-task learning; when the tasks are substantially different from each other, our method can still obtain competitive convergence rates compared to single-task learning. Similarly for TL, our method can achieve better upper bounds than those from fitting GMM only to target data when the target and sources are similar, and remains competitive otherwise. In addition, the derived upper bounds reveal the robustness of our methods against a fraction of outlier tasks (for MTL) or outlier sources (for TL) from arbitrary distributions. These guarantees certify our procedures as adaptive (to the unknown task relatedness) and robust (to contaminated data) learning approaches.
(iii) We derive the minimax lower bounds for parameter estimation and excess mis-clustering errors. In various regimes, the upper bounds from our methods match the lower bounds (up to small order terms), showing that the proposed methods are (nearly) minimax rate optimal.
(iv) Our MTL and TL approaches require the initial estimates for different tasks to be "well-aligned", due to the non-identifiability of GMM. We propose two pre-processing alignment algorithms to provably resolve the alignment problem. Similar problems arise in many unsupervised MTL settings. However, to the best of our knowledge, there is no formal discussion of the alignment issue in the existing literature of unsupervised MTL (Gu, Li, and Han 2011; Zhang and Zhang 2011; Yang et al. 2014; Zhang et al. 2018; Dieuleveut et al. 2021; Marfoq et al. 2021). Therefore, our rigorous treatment of the alignment problem is an important step forward in this field.

The rest of the article is organized as follows. In Section 2, we first discuss the multi-task learning problem for binary GMMs, by introducing the problem setting, our method, and the associated theory. The above-mentioned alignment problem is discussed in Section 2.4. We present a simulation study and a real-data analysis in Section 3 to validate our theoretical findings. Finally, in Section 4, we highlight several interesting extensions that are deferred to the supplementary materials due to space constraints. These include the extension to multi-cluster GMMs, discussions on handling different numbers of clusters across tasks, determination of cluster numbers, and modeling heterogeneous covariance matrices across clusters. Additional numerical results, a complete treatment of the transfer learning setting, and all technical proofs are also provided in the supplementary materials.

We summarize the notations used throughout the article here for convenience. We use bold capital letters (e.g., $\boldsymbol{\Sigma}$) to denote matrices and bold small letters (e.g., $\boldsymbol{x}$, $\boldsymbol{y}$) to denote vectors. For a matrix $\boldsymbol{A} = [a_{ij}]_{p\times q} \in \mathbb{R}^{p\times q}$, its 2-norm or spectral norm is defined as $\|\boldsymbol{A}\|_2 = \max_{\boldsymbol{x}:\|\boldsymbol{x}\|_2=1} \|\boldsymbol{A}\boldsymbol{x}\|_2$. If $q = 1$, $\boldsymbol{A}$ becomes a p-dimensional vector and $\|\boldsymbol{A}\|_2$ equals its Euclidean norm. For symmetric $\boldsymbol{A}$, we define $\lambda_{\max}(\boldsymbol{A})$ and $\lambda_{\min}(\boldsymbol{A})$ as the maximum and minimum eigenvalues of $\boldsymbol{A}$, respectively. For two nonzero real sequences $\{a_n\}_{n=1}^\infty$ and $\{b_n\}_{n=1}^\infty$, we use $a_n \ll b_n$, $b_n \gg a_n$ or $a_n = \mathcal{o}(b_n)$ to represent $|a_n/b_n| \to 0$ as $n \to \infty$. And $a_n \lesssim b_n$, $b_n \gtrsim a_n$ or $a_n = \mathcal{O}(b_n)$ means $\sup_n |a_n/b_n| < \infty$. For two random variable sequences $\{x_n\}_{n=1}^\infty$ and $\{y_n\}_{n=1}^\infty$, the notation $x_n = \mathcal{O}_{\mathbb{P}}(y_n)$ means that for any $\epsilon > 0$, there exists a positive constant M such that $\sup_n \mathbb{P}(|x_n/y_n| > M) \le \epsilon$. For two real numbers a and b, $a \vee b$ and $a \wedge b$ represent $\max(a, b)$ and $\min(a, b)$, respectively. For any positive integer K, both $1 : K$ and $[K]$ stand for the set $\{1, 2, \dots, K\}$. For any set $S \subseteq [K]$, $|S|$ denotes its cardinality, and S^c denotes its complement. Unless otherwise specified, c, C, C_1, C_2, ... represent some positive constants and can change from line to line.

2. Multi-Task Learning

2.1. Problem Setting

Suppose there are K tasks, for which we have n_k observations $\{\boldsymbol{z}_i^{(k)}\}_{i=1}^{n_k}$ from the kth task. Suppose there exists an *unknown* subset $S \subseteq 1 : K$, such that observations from each task in S independently follow a GMM, while samples from tasks outside S can be arbitrarily distributed. This means,

$$y_i^{(k)} = \begin{cases} 1, & \text{with probability } 1 - w^{(k)*}; \\ 2, & \text{with probability } w^{(k)*}; \end{cases}$$
$$\boldsymbol{z}_i^{(k)}|y_i^{(k)} = r \sim \mathcal{N}(\boldsymbol{\mu}_r^{(k)*}, \boldsymbol{\Sigma}^{(k)*}), \quad r = 1, 2,$$

for all $k \in S$, $i = 1 : n_k$, and

$$\{\boldsymbol{z}_i^{(k)}\}_{i,k\in S^c} \sim \mathbb{Q}_S,$$

where $\mathbb{Q}_S$ is some probability measure on $(\mathbb{R}^p)^{\otimes n_{S^c}}$ and $n_{S^c} = \sum_{k\in S^c} n_k$. In unsupervised learning, we have no access to the true labels $\{y_i^{(k)}\}_{i,k}$. To formalize the multi-task learning problem, we first introduce the parameter space for a single GMM:

$$\begin{aligned}\overline{\Theta} = \{\overline{\boldsymbol{\theta}} = (w, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}) :\ & \|\boldsymbol{\mu}_1\|_2 \vee \|\boldsymbol{\mu}_2\|_2 \le M, w \in (c_w, 1 - c_w), \\ & c_{\boldsymbol{\Sigma}}^{-1} \le \lambda_{\min}(\boldsymbol{\Sigma}) \le \lambda_{\max}(\boldsymbol{\Sigma}) \le c_{\boldsymbol{\Sigma}}\},\end{aligned}$$

where M, $c_w \in (0, 1/2]$ and $c_{\boldsymbol{\Sigma}}$ are some fixed positive constants. For simplicity, throughout the main text, we assume these constants are fixed. Hence, we have suppressed the dependency on them in the notation $\overline{\Theta}$. The parameter space $\overline{\Theta}$ is a standard formulation. Similar parameter spaces have been considered, for example, in Cai, Ma, and Zhang (2019).

Our goal of multi-task learning is to leverage the potential similarity shared by different tasks in S to collectively learn them. The tasks outside S can be arbitrarily distributed, and they can potentially be outlier tasks. This motivates us to define a joint parameter space for GMMs in S:

$$\overline{\Theta}_S(h) = \Big\{ \{\overline{\boldsymbol{\theta}}^{(k)}\}_{k\in S} = \{(w^{(k)}, \boldsymbol{\mu}_1^{(k)}, \boldsymbol{\mu}_2^{(k)}, \boldsymbol{\Sigma}^{(k)})\}_{k\in S} : \overline{\boldsymbol{\theta}}^{(k)} \in \overline{\Theta}, \inf_{\overline{\boldsymbol{\beta}}} \max_{k\in S} \|\boldsymbol{\beta}^{(k)} - \overline{\boldsymbol{\beta}}\|_2 \le h \Big\}, \tag{2}$$

where $\boldsymbol{\beta}^{(k)} = (\boldsymbol{\Sigma}^{(k)})^{-1}(\boldsymbol{\mu}_2^{(k)} - \boldsymbol{\mu}_1^{(k)})$ is called the discriminant coefficient in the kth task (recall Section 1.1). For convenience, we define $\delta^{(k)} = (\boldsymbol{\beta}^{(k)})^\top(\boldsymbol{\mu}_1^{(k)} + \boldsymbol{\mu}_2^{(k)})/2$, which together with $\log((1 - w^{(k)})/w^{(k)})$ is part of the decision boundary. Note that this parameter space is defined only for GMMs of tasks in S. To model potentially corrupted or contaminated data, we do not impose any distributional constraints for tasks in S^c. Such a modeling framework is reminiscent of Huber's ϵ-contamination model (Huber 1964). Similar formulations have been adopted in recent multi-task learning research, such as Konstantinov et al. (2020) and Duan and Wang (2023).

For GMMs in $\overline{\Theta}_S(h)$, we assume that they share similar discriminant coefficients. The similarity is formalized by assuming that all the discriminant coefficients in S are within Euclidean distance h from a "center". Given that the discriminant coefficient has a major impact on the clustering performance (see the discriminant rule in (1)), the parameter space $\overline{\Theta}_S(h)$ is tailored to characterize the task relatedness from the clustering perspective. A similar viewpoint that focuses on modeling the discriminant coefficient has appeared in the study of high-dimensional GMM clustering (Cai, Ma, and Zhang 2019) and sparse linear discriminant analysis (Cai and Liu 2011; Mai, Zou, and Yuan 2012). With both S and h being unknown in practice, we aim to develop a multi-task learning procedure that is robust to outlier tasks in S^c, and achieves improved performance for tasks in S (compared to the single-task learning), in terms of discriminant coefficient estimation and clustering, whenever h is small.

The parameter space does not require the mean vectors $\{\boldsymbol{\mu}_1^{(k)}, \boldsymbol{\mu}_2^{(k)}\}_{k\in S}$ or the covariance matrices $\{\boldsymbol{\Sigma}^{(k)}\}_{k\in S}$ to be similar, although they are not free parameters due to the constraint on $\{\boldsymbol{\beta}^{(k)}\}_{k\in S}$. And the mixture proportions $\{w^{(k)}\}_{k\in S}$ do not need to be similar either. We thus avoid imposing restrictive conditions on those parameters. On the other hand, it implies that estimation of the mixture proportions, mean vectors, and covariance matrices in multi-task learning may not be generally improvable over that in single-task learning. This is verified by the theoretical results in Section 2.3. While the current treatment in the article does not consider similarity structure among $\{\boldsymbol{\mu}_1^{(k)}, \boldsymbol{\mu}_2^{(k)}\}_{k\in S}$, $\{\boldsymbol{\Sigma}^{(k)}\}_{k\in S}$ or $\{w^{(k)}\}_{k\in S}$, our methods and theory can be readily adapted to handle such scenarios, if desired.

There are two main reasons why this MTL problem can be challenging. First, commonly used strategies like data pooling are fragile with respect to outlier tasks and can lead to arbitrarily inaccurate outcomes in the presence of even a small number of outliers. Also, since the distribution of data from outlier tasks can be adversarial to the learner, the idea of outlier task detection in the recent literature (Li, Cai, and Li 2021; Tian and Feng 2023) may not be applicable. Second, to address the nonconvexity of the likelihood, we propose to explore the similarity among tasks via a generalization of the EM algorithm. However, a clear theoretical understanding of such an iterative procedure requires a delicate analysis of the whole iterative process. In particular, as in the analysis of EM algorithms (Cai, Ma, and Zhang 2019; Kwon and Caramanis 2020), the estimates of similar discriminant vectors $\{\boldsymbol{\beta}^{(k)*}\}_{k\in S}$ and other potentially dissimilar parameters are entangled in the iterations. It is highly nontrivial to separate the impact of estimating $\{\boldsymbol{\beta}^{(k)*}\}_{k\in S}$ and other parameters to derive the desired statistical error rates. We address this challenge through a localization technique by carefully shrinking the analysis radius of estimators as the iteration proceeds.

2.2. Method

We aim to tackle the problem of GMM estimation under the context of multi-task learning. The EM algorithm is commonly used to address the non-convexity of the log-likelihood function arising from the latent labels. In the standard EM algorithm, we "classify" the observations (update the posterior) in E-step and update the parameter estimations in M-step (Redner and Walker 1984). For multi-task and transfer learning problems, the penalization framework is very popular, where we solve an optimization problem based on a new objective function. This objective function consists of a local loss function and a penalty term, forcing the estimators of similar tasks to be close to each other. For example, see Zhang and Zhang (2011), Zhang, Zhang, and Liu (2015), Bastani (2021), Xu and Bastani (2021), Li, Cai, and Li (2021), Duan and Wang (2023), Lin and Reimherr (2022), Li, Cai, and Duan (2023), and Tian and Feng (2023). Thus motivated, our method seeks a combination of the EM algorithm and the penalization framework.

In particular, we adapt the penalization framework of Duan and Wang (2023) and modify the updating formulas in the M-step accordingly. The proposed procedure is summarized in Algorithm 1. For simplicity, in Algorithm 1 we have used the notation

$$\gamma_{\boldsymbol{\theta}}(\boldsymbol{z}) = \frac{w\exp(\boldsymbol{\beta}^\top\boldsymbol{z} - \delta)}{1 - w + w\exp(\boldsymbol{\beta}^\top\boldsymbol{z} - \delta)}, \quad \text{for } \boldsymbol{\theta} = (w, \boldsymbol{\beta}, \delta).$$

Note that $\gamma_{\boldsymbol{\theta}}(\boldsymbol{z})$ is the posterior probability $\mathbb{P}(Y = 2|Z = \boldsymbol{z})$ given the observation $\boldsymbol{z}$. The estimated posterior probability is calculated in every E-step, given the updated parameter estimates.

Recall that the parameter space $\overline{\Theta}_S(h)$ introduced in (2) does not encode similarity for the mixture proportions $\{w^{(k)*}\}_{k\in S}$, mean vectors $\{\boldsymbol{\mu}_1^{(k)*}, \boldsymbol{\mu}_2^{(k)*}\}_{k\in S}$, or covariance matrices $\{\boldsymbol{\Sigma}^{(k)*}\}_{k\in S}$. Hence, their updates in Steps 5–7 are kept the same as those in the standard EM algorithm. Regarding the update for discriminant coefficients in Step 9, the quadratic loss function is motivated by the direct estimation of the discriminant coefficient in high-dimensional GMM (Cai, Ma, and Zhang 2019) and high-dimensional LDA literature (Cai and Liu 2011; Witten and Tibshirani 2011; Fan, Feng, and Tong 2012; Mai, Zou, and Yuan 2012; Mai, Yang, and Zou 2019). The penalty term in Step 9 penalizes the contrasts

Algorithm 1: MTL-GMM

Input: Initialization $\{(\widehat{w}^{(k)[0]}, \widehat{\boldsymbol{\beta}}^{(k)[0]}, \widehat{\boldsymbol{\mu}}_1^{(k)[0]}, \widehat{\boldsymbol{\mu}}_2^{(k)[0]})\}_{k=1}^K$, maximum number of iteration rounds T, initial penalty parameter $\lambda^{[0]}$, tuning parameters $C_\lambda > 0$ and $\kappa \in (0,1)$

1 $\widehat{\boldsymbol{\theta}}^{(k)[0]} = (\widehat{w}^{(k)[0]}, \widehat{\boldsymbol{\beta}}^{(k)[0]}, \widehat{\delta}^{(k)[0]})$ and $\widehat{\delta}^{(k)[0]} = \frac{1}{2}(\widehat{\boldsymbol{\beta}}^{(k)[0]})^\top(\widehat{\boldsymbol{\mu}}_1^{(k)[0]} + \widehat{\boldsymbol{\mu}}_2^{(k)[0]})$ for $k = 1 : K$

2 **for** $t = 1$ **to** T **do**

3 $\quad \lambda^{[t]} = \kappa\lambda^{[t-1]} + C_\lambda\sqrt{p + \log K}$ `// Penalty parameter update`

4 $\quad$ **for** $k = 1$ **to** K **do** `// Local update for each task`

5 $\quad\quad \widehat{w}^{(k)[t]} = \frac{1}{n_k}\sum_{i=1}^{n_k}\gamma_{\widehat{\boldsymbol{\theta}}^{(k)[t-1]}}(\boldsymbol{z}_i^{(k)})$

6 $\quad\quad \widehat{\boldsymbol{\mu}}_1^{(k)[t]} = \frac{\sum_{i=1}^{n_k}[1-\gamma_{\widehat{\boldsymbol{\theta}}^{(k)[t-1]}}(\boldsymbol{z}_i^{(k)})]\boldsymbol{z}_i^{(k)}}{n_k(1-\widehat{w}^{(k)[t]})}$, $\widehat{\boldsymbol{\mu}}_2^{(k)[t]} = \frac{\sum_{i=1}^{n_k}\gamma_{\widehat{\boldsymbol{\theta}}^{(k)[t-1]}}(\boldsymbol{z}_i^{(k)})\boldsymbol{z}_i^{(k)}}{n_k\widehat{w}^{(k)[t]}}$

7 $\quad\quad \widehat{\boldsymbol{\Sigma}}^{(k)[t]} = \frac{1}{n_k}\sum_{i=1}^{n_k}\Big\{[1-\gamma_{\widehat{\boldsymbol{\theta}}^{(k)[t-1]}}(\boldsymbol{z}_i^{(k)})]\cdot(\boldsymbol{z}_i^{(k)} - \widehat{\boldsymbol{\mu}}_1^{(k)[t]})(\boldsymbol{z}_i^{(k)} - \widehat{\boldsymbol{\mu}}_1^{(k)[t]})^\top + \gamma_{\widehat{\boldsymbol{\theta}}^{(k)[t-1]}}(\boldsymbol{z}_i^{(k)})\cdot(\boldsymbol{z}_i^{(k)} - \widehat{\boldsymbol{\mu}}_2^{(k)[t]})(\boldsymbol{z}_i^{(k)} - \widehat{\boldsymbol{\mu}}_2^{(k)[t]})^\top\Big\}$

8 $\quad$ **end**

9 $\quad \{\widehat{\boldsymbol{\beta}}^{(k)[t]}\}_{k=1}^K$, $\overline{\boldsymbol{\beta}}^{[t]} = \underset{\boldsymbol{\beta}^{(1)},\ldots,\boldsymbol{\beta}^{(K)},\overline{\boldsymbol{\beta}}}{\arg\min}\Big\{\sum_{k=1}^K n_k\Big[\frac{1}{2}(\boldsymbol{\beta}^{(k)})^\top\widehat{\boldsymbol{\Sigma}}^{(k)[t]}\boldsymbol{\beta}^{(k)} - (\boldsymbol{\beta}^{(k)})^\top(\widehat{\boldsymbol{\mu}}_2^{(k)[t]} - \widehat{\boldsymbol{\mu}}_1^{(k)[t]})\Big] + \sum_{k=1}^K\sqrt{n_k}\lambda^{[t]}\cdot\|\boldsymbol{\beta}^{(k)} - \overline{\boldsymbol{\beta}}\|_2\Big\}$

$\quad$ `// Aggregation to learn` $\{\widehat{\boldsymbol{\beta}}^{(k)[t]}\}_{k=1}^K$

10 $\quad$ **for** $k = 1$ **to** K **do** `// Local update for each task`

11 $\quad\quad \widehat{\delta}^{(k)[t]} = \frac{1}{2}(\widehat{\boldsymbol{\beta}}^{(k)[t]})^\top(\widehat{\boldsymbol{\mu}}_1^{(k)[t]} + \widehat{\boldsymbol{\mu}}_2^{(k)[t]})$

12 $\quad\quad$ Let $\widehat{\boldsymbol{\theta}}^{(k)[t]} = (\widehat{w}^{(k)[t]}, \widehat{\boldsymbol{\beta}}^{(k)[t]}, \widehat{\delta}^{(k)[t]})$

13 $\quad$ **end**

14 **end**

Output: $\{(\widehat{\boldsymbol{\theta}}^{(k)[T]}, \widehat{\boldsymbol{\mu}}_1^{(k)[T]}, \widehat{\boldsymbol{\mu}}_2^{(k)[T]}, \widehat{\boldsymbol{\Sigma}}^{(k)[T]})\}_{k=1}^K$ with $\widehat{\boldsymbol{\theta}}^{(k)[T]} = (\widehat{w}^{(k)[T]}, \widehat{\boldsymbol{\beta}}^{(k)[T]}, \widehat{\delta}^{(k)[T]})$, and $\overline{\boldsymbol{\beta}}^{[T]}$

of $\boldsymbol{\beta}^{(k)}$'s to exploit the similarity structure among tasks. The "center" parameter $\overline{\boldsymbol{\beta}}$ in the penalization induces robustness against outlier tasks. We refer to Duan and Wang (2023) for a systematic treatment of this penalization framework. It is straightforward to verify that when the tuning parameters $\{\lambda^{[t]}\}_{t=1}^T$ are set to zero, Algorithm 1 reduces to the standard EM algorithm performed separately on the K tasks. That is, for each $k = 1 : K$, given the parameter estimate from the previous step $\widehat{\boldsymbol{\theta}}^{(k)[t-1]} = (\widehat{w}^{(k)[t-1]}, \widehat{\boldsymbol{\beta}}^{(k)[t-1]}, \widehat{\delta}^{(k)[t-1]})$, we update $\widehat{w}^{(k)[t]}$, $\widehat{\boldsymbol{\mu}}_1^{(k)[t]}$, $\widehat{\boldsymbol{\mu}}_2^{(k)[t]}$, $\widehat{\boldsymbol{\Sigma}}^{(k)[t]}$, and $\widehat{\delta}^{(k)[t]}$ as in Algorithm 1, and update $\widehat{\boldsymbol{\beta}}^{(k)[t]}$ via

$$\widehat{\boldsymbol{\beta}}^{(k)[t]} = (\widehat{\boldsymbol{\Sigma}}^{(k)[t]})^{-1}(\widehat{\boldsymbol{\mu}}_2^{(k)[t]} - \widehat{\boldsymbol{\mu}}_1^{(k)[t]}).$$

For the maximum number of iteration rounds, T, our theory will show that $T \gtrsim \log(\max_{k=1:K} n_k)$ is sufficient to reach the desired statistical error rates. In practice, we can terminate the iteration when the change of estimates between two successive rounds falls below some pre-set small tolerance level. We discuss the initialization in detail in Sections 2.3 and 2.4.

Lastly, it is worth noting that the aggregation in Step 9 is central to Algorithm 1. The joint optimization over the individual parameters $\{\boldsymbol{\beta}^{(k)}\}_{k=1}^K$ and a global parameter $\overline{\boldsymbol{\beta}}$ using an ℓ_2-penalty provides sufficient flexibility for multi-task learning. As we will show later, with appropriate choices of the penalty parameter $\lambda^{[t]}$, Step 9 enables adaptive information sharing: when the $\boldsymbol{\beta}^{(k)*}$'s are sufficiently similar, the penalty reduces estimation variance and encourages the estimators $\{\boldsymbol{\beta}^{(k)}\}_{k=1}^K$ to be close to one another; when the $\boldsymbol{\beta}^{(k)*}$'s differ significantly, the strongly convex squared loss dominates the penalty term, ensuring that the estimators remain close to their corresponding single-task solutions. Moreover, as discussed in She and Owen (2011) and Donoho and Montanari (2016), this form of penalization is connected to robust empirical risk minimization. Specifically, the equivalent loss function (i.e., the objective function in Step 9 after profiling out $\{\boldsymbol{\beta}^{(k)}\}_{k=1}^K$) for estimating the global parameter $\overline{\boldsymbol{\beta}}$ can be shown to have some robustness properties, providing robustness against a small proportion of outlier tasks.

2.3. Theory

In this section, we develop statistical theories for our proposed procedure MTL-GMM (see Algorithm 1). As mentioned in Section 2.1, we are interested in the performance of both parameter estimation and clustering, although the latter is the main focus and motivation. First, we impose conditions in the following assumption set.

Assumption 1. Denote

$$\Delta^{(k)}=\sqrt{(\boldsymbol{\mu}_1^{(k)*}-\boldsymbol{\mu}_2^{(k)*})^\top(\boldsymbol{\Sigma}^{(k)*})^{-1}(\boldsymbol{\mu}_1^{(k)*}-\boldsymbol{\mu}_2^{(k)*})}$$

for $k\in S$. The quantity $\Delta^{(k)}$ is the Mahalanobis distance between $\boldsymbol{\mu}_1^{(k)*}$ and $\boldsymbol{\mu}_2^{(k)*}$ with covariance matrix $\boldsymbol{\Sigma}^{(k)*}$, and can be viewed as the signal-to-noise ratio (SNR) in the kth task (Anderson 1958). Suppose the following conditions hold:

(i) $n_S=\sum_{k\in S}n_k\ge C_1|S|\max_{k=1:K}n_k$ with a constant $C_1\in(0,1]$;
(ii) $\min_{k\in S}n_k\ge C_2(p+\log K)$ with some constant $C_2>0$;
(iii) Either of the following two conditions holds with some constant $C_3>0$:
 (a) $\max_{k\in S}\left(\|\widehat{\boldsymbol{\beta}}^{(k)[0]}-\boldsymbol{\beta}^{(k)*}\|_2\vee\|\widehat{\boldsymbol{\mu}}_1^{(k)[0]}-\boldsymbol{\mu}_1^{(k)*}\|_2\vee\|\widehat{\boldsymbol{\mu}}_2^{(k)[0]}-\boldsymbol{\mu}_2^{(k)*}\|_2\right)\le C_3\min_{k\in S}\Delta^{(k)}$, $\max_{k\in S}|\widehat{w}^{(k)[0]}-w^{(k)*}|\le c_w/2$;
 (b) $\max_{k\in S}\left(\|\widehat{\boldsymbol{\beta}}^{(k)[0]}+\boldsymbol{\beta}^{(k)*}\|_2\vee\|\widehat{\boldsymbol{\mu}}_1^{(k)[0]}-\boldsymbol{\mu}_2^{(k)*}\|_2\vee\|\widehat{\boldsymbol{\mu}}_2^{(k)[0]}-\boldsymbol{\mu}_1^{(k)*}\|_2\right)\le C_3\min_{k\in S}\Delta^{(k)}$, $\max_{k\in S}|1-\widehat{w}^{(k)[0]}-w^{(k)*}|\le c_w/2$.
(iv) $\min_{k\in S}\Delta^{(k)}\ge C_4>0$ with some constant $C_4>0$;

Remark 1. These are common and mild conditions related to the sample size, initialization, and signal-to-noise ratio of GMMs. Condition (i) requires the maximum sample size of all tasks not to be much larger than the average sample size of tasks in S. Similar conditions can be found in Duan and Wang (2023). Condition (ii) is the requirement on the sample size of tasks in S. The usual condition for low-dimensional single-task learning is $n_k\gtrsim p$ (Cai, Ma, and Zhang 2019). The additional $\log K$ term arises from the simultaneous control of performance on all tasks in S, where S can be as large as $1:K$. Condition (iii) requires that the initialization should not be too far away from the truth, which is commonly assumed in either the analysis of EM algorithm (Redner and Walker 1984; Balakrishnan, Wainwright, and Yu 2017; Cai, Ma, and Zhang 2019) or other iterative algorithms like the local estimation used in semi-parametric models (Carroll et al. 1997; Li and Liang 2008) and adaptive Lasso (Zou 2006). The two possible forms considered in this condition are due to the fact that binary GMM is only identifiable up to label permutation. Condition (iv) requires that the signal strength of GMM (in terms of Mahalanobis distance) is strong enough, which is usually assumed in the literature about the likelihood-based methods of GMMs (Dasgupta and Schulman 2000; Azizyan, Singh, and Wasserman 2013; Balakrishnan, Wainwright, and Yu 2017; Cai, Ma, and Zhang 2019).

We first establish the rate of convergence for the estimation. Recalling the parameter space $\overline{\Theta}_S(h)$ in (2), let us denote the true parameter by

$$\{\overline{\boldsymbol{\theta}}^{(k)*}\}_{k\in S}=\{(w^{(k)*},\boldsymbol{\mu}_1^{(k)*},\boldsymbol{\mu}_2^{(k)*},\boldsymbol{\Sigma}^{(k)*})\}_{k\in S}\in\overline{\Theta}_S(h).$$

To better present the results for parameters related to the optimal discriminant rule (1), we further denote

$$\boldsymbol{\theta}^{(k)*}=(w^{(k)*},\boldsymbol{\beta}^{(k)*},\delta^{(k)*}),\quad \forall k\in S,$$

where $\boldsymbol{\beta}^{(k)*}=(\boldsymbol{\Sigma}^{(k)*})^{-1}(\boldsymbol{\mu}_2^{(k)*}-\boldsymbol{\mu}_1^{(k)*})$, $\delta^{(k)*}=\frac{1}{2}(\boldsymbol{\beta}^{(k)*})^\top(\boldsymbol{\mu}_1^{(k)*}+\boldsymbol{\mu}_2^{(k)*})$. Note that $\boldsymbol{\theta}^{(k)*}$ is a function of $\overline{\boldsymbol{\theta}}^{(k)*}$. For the estimators returned by MTL-GMM (see Algorithm 1), we are particularly interested in the following two error metrics:

$$\begin{aligned}
&d(\widehat{\boldsymbol{\theta}}^{(k)[T]},\boldsymbol{\theta}^{(k)*})\\
&:=\min\{|\widehat{w}^{(k)[T]}-w^{(k)*}|\vee\|\widehat{\boldsymbol{\beta}}^{(k)[T]}-\boldsymbol{\beta}^{(k)*}\|_2\\
&\quad\vee|\widehat{\delta}^{(k)[T]}-\delta^{(k)*}|,\\
&|1-\widehat{w}^{(k)[T]}-w^{(k)*}|\vee\|\widehat{\boldsymbol{\beta}}^{(k)[T]}+\boldsymbol{\beta}^{(k)*}\|_2\\
&\quad\vee|\widehat{\delta}^{(k)[T]}+\delta^{(k)*}|\},\\
&\left(\min_{\pi:[2]\to[2]}\max_{r=1:2}\|\widehat{\boldsymbol{\mu}}_r^{(k)[T]}-\boldsymbol{\mu}_{\pi(r)}^{(k)*}\|_2\right)\vee\|\widehat{\boldsymbol{\Sigma}}^{(k)[T]}-\boldsymbol{\Sigma}^{(k)*}\|_2,
\end{aligned}$$

where $\pi:[2]\to[2]$ is a permutation on $\{1,2\}$. Again, we take the minimum above because binary GMM is identifiable up to label permutation. The first error metric $d(\widehat{\boldsymbol{\theta}}^{(k)[T]},\boldsymbol{\theta}^{(k)*})$ involves the error for discriminant coefficients and is closely related to the clustering performance. It reveals how well our method uses similarity structure in multi-task learning. The second error metric is about the mean vectors and covariance matrix. As discussed in Section 2.1, we shall not expect it to be improved compared to that in single-task learning, as these parameters are not necessarily similar.

We are ready to present upper bounds for the estimation error of MTL-GMM. We recall that $\overline{\Theta}_S(h)$ and $\mathbb{Q}_S$ are the parameter space and probability measure that we use in Section 2.1 to describe the data distributions for tasks in S and S^c, respectively.

Theorem 1 (Upper bounds of the estimation error of GMM parameters for MTL-GMM). Suppose Assumption 1 holds for some S with $|S|\ge s$ and $\epsilon:=\frac{K-s}{K}<1/3$. Let $\lambda^{[0]}\ge C_1\max_{k=1:K}\sqrt{n_k}$, $C_\lambda\ge C_1$ and $\kappa>C_2$ with some constants $C_1>0, C_2\in(0,1)$ [2]. Then there exists a constant $C_3>0$, such that for any $\{\overline{\boldsymbol{\theta}}^{(k)*}\}_{k\in S}=\{(w^{(k)*},\boldsymbol{\mu}_1^{(k)*},\boldsymbol{\mu}_2^{(k)*},\boldsymbol{\Sigma}^{(k)*})\}_{k\in S}\in\overline{\Theta}_S(h)$ and any probability measure $\mathbb{Q}_S$ on $(\mathbb{R}^p)^{\otimes n_{S^c}}$, with probability $1-C_3K^{-1}$, the following hold for all $k\in S$:

$$\begin{aligned}
d(\widehat{\boldsymbol{\theta}}^{(k)[T]},\boldsymbol{\theta}^{(k)*})&\lesssim\sqrt{\frac{p}{n_S}}+\sqrt{\frac{\log K}{n_k}}+h\wedge\sqrt{\frac{p+\log K}{n_k}}\\
&\quad+\epsilon\sqrt{\frac{p+\log K}{\max_{k=1:K}n_k}}+T^2(\kappa')^T,\\
&\left(\min_{\pi:[2]\to[2]}\max_{r=1:2}\|\widehat{\boldsymbol{\mu}}_r^{(k)[T]}-\boldsymbol{\mu}_{\pi(r)}^{(k)*}\|_2\right)\\
&\quad\vee\|\widehat{\boldsymbol{\Sigma}}^{(k)[T]}-\boldsymbol{\Sigma}^{(k)*}\|_2\lesssim\sqrt{\frac{p+\log K}{n_k}}+T^2(\kappa')^T,
\end{aligned}$$

where $\kappa'\in(0,1)$ is some constant and $n_S=\sum_{k\in S}n_k$. When $T\ge C\log(\max_{k=1:K}n_k)$ with a large constant $C>0$, the last term on the right-hand side will be dominated by other terms in both inequalities.

The upper bound of $\left(\min_{\pi:[2]\to[2]}\max_{r=1:2}\|\widehat{\boldsymbol{\mu}}_r^{(k)[T]}-\boldsymbol{\mu}_{\pi(r)}^{(k)*}\|_2\right)\vee\|\widehat{\boldsymbol{\Sigma}}^{(k)[T]}-\boldsymbol{\Sigma}^{(k)*}\|_2$ contains two parts. The first

[2] C_1 and C_2 depend on the constants M, c_w, and $c_{\boldsymbol{\Sigma}}$ etc.

part is comparable to the single-task learning rate (Cai, Ma, and Zhang 2019) (up to a $\sqrt{\log K}$ term due to the simultaneous control over all tasks in S), and the second part characterizes the geometric convergence of iterates. As expected, since $\boldsymbol{\mu}_1^{(k)*}$, $\boldsymbol{\mu}_2^{(k)*}$, $\boldsymbol{\Sigma}^{(k)*}$ in S are not necessarily similar, an improved error rate over single-task learning is generally impossible. The upper bound for $d(\widehat{\boldsymbol{\theta}}^{(k)[T]}, \boldsymbol{\theta}^{(k)*})$ is directly related to the clustering performance of our method. Thus, we will provide a detailed discussion of it after presenting the clustering result in the next theorem.

As introduced in Section 1.1, using the estimate $\widehat{\boldsymbol{\theta}}^{(k)[T]} = (\widehat{w}^{(k)[T]}, \widehat{\boldsymbol{\beta}}^{(k)[T]}, \widehat{\delta}^{(k)[T]})$ from Algorithm 1, we can construct a classifier for task k as

$$\widehat{\mathcal{C}}^{(k)[T]}(\boldsymbol{z}) = \begin{cases} 1, & \text{if } (\widehat{\boldsymbol{\beta}}^{(k)[T]})^\top \boldsymbol{z} - \widehat{\delta}^{(k)[T]} \le \log\left(\frac{1-\widehat{w}^{(k)[T]}}{\widehat{w}^{(k)[T]}}\right); \\ 2, & \text{otherwise.} \end{cases}$$

Recall that for a clustering method $\mathcal{C} : \mathbb{R}^p \to \{1,2\}$, its mis-clustering error rate under GMM with parameter $\overline{\boldsymbol{\theta}} = (w, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma})$ is

$$R_{\overline{\boldsymbol{\theta}}}(\mathcal{C}) = \min_{\pi:[2]\to[2]} \mathbb{P}_{\overline{\boldsymbol{\theta}}}(\mathcal{C}(Z^{\text{new}}) \neq \pi(Y^{\text{new}})),$$

where $Z^{\text{new}} \sim (1-w)\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}) + w\mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$ is a future observation associated with the label Y^{new}, independent from $\mathcal{C}$; the probability $\mathbb{P}_{\overline{\boldsymbol{\theta}}}$ is w.r.t. $(Z^{\text{new}}, Y^{\text{new}})$, and the minimum is taken over two permutation functions on $\{1,2\}$. Denote $\mathcal{C}_{\overline{\boldsymbol{\theta}}}$ as the Bayes classifier that minimizes $R_{\overline{\boldsymbol{\theta}}}(\mathcal{C})$. In the following theorem, we obtain the upper bound of the excess mis-clustering error of $\widehat{\mathcal{C}}^{(k)[T]}$ for $k \in S$.

Theorem 2 (Upper bound of the excess mis-clustering error for MTL-GMM). Suppose the same conditions in Theorem 1 hold. Then there exists a constant $C_1 > 0$ such that for any $\{\overline{\boldsymbol{\theta}}^{(k)*}\}_{k\in S} \in \overline{\Theta}_S(h)$ and any probability measure $\mathbb{Q}_S$ on $(\mathbb{R}^p)^{\otimes n_{S^c}}$, with probability $1 - C_1K^{-1}$, the following holds for all $k \in S$:

$$\begin{aligned} &R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\widehat{\mathcal{C}}^{(k)[T]}) - R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\mathcal{C}_{\overline{\boldsymbol{\theta}}^{(k)*}}) \\ &\lesssim d^2(\widehat{\boldsymbol{\theta}}^{(k)[T]}, \boldsymbol{\theta}^{(k)*}) \\ &\lesssim \underbrace{\frac{p}{n_S}}_{\text{(I)}} + \underbrace{\frac{\log K}{n_k}}_{\text{(II)}} + \underbrace{h^2 \wedge \frac{p+\log K}{n_k}}_{\text{(III)}} \\ &\quad + \underbrace{\epsilon^2 \frac{p+\log K}{\max_{k=1:K} n_k}}_{\text{(IV)}} + \underbrace{T^4(\kappa')^{2T}}_{\text{(V)}}, \end{aligned}$$

with some $\kappa' \in (0,1)$. When $T \ge C\log(\max_{k=1:K} n_k)$ with a large constant $C > 0$, the last term on the right-hand side will be dominated by the second term.

The upper bounds of $d(\widehat{\boldsymbol{\theta}}^{(k)[T]}, \boldsymbol{\theta}^{(k)*})$ in Theorem 1 and $R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\widehat{\mathcal{C}}^{(k)[T]}) - R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\mathcal{C}_{\overline{\boldsymbol{\theta}}^{(k)*}})$ in Theorem 2 consist of five parts with one-to-one correspondence. It is sufficient to discuss the bound of $R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\widehat{\mathcal{C}}^{(k)[T]}) - R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\mathcal{C}_{\overline{\boldsymbol{\theta}}^{(k)*}})$. Part (I) represents the "oracle rate," which can be achieved when all tasks in S are the same. This is the best rate to achieve possibly. Part (II) is a dimension-free error caused by estimating scalar parameters $\delta^{(k)*}$ and $w^{(k)*}$ that appears in the optimal discriminant rule. Part (III) includes h that measures the degree of similarity among the tasks in S. When these tasks are very similar, h will be small, contributing a small term to the upper bound. Notably, even when h is large, the term becomes $\frac{p+\log K}{n_k}$, and it is still comparable to the minimax error rate of single-task learning $\mathcal{O}_{\mathbb{P}}(p/n_k)$ (e.g., Theorems 4.1 and 4.2 in Cai, Ma, and Zhang 2019). We have the extra $\log K$ term here due to the simultaneous control over all tasks in S. Part (IV) quantifies the influence from the outlier tasks in S^c. When there are more outlier tasks, ϵ increases, and the bound becomes worse. On the other hand, as long as ϵ is small enough to make this term dominated by any other part, the error rate induced by outlier tasks becomes negligible. Given that data from outlier tasks can be arbitrarily contaminated, we can conclude that our method is robust against a fraction of outlier tasks from arbitrary sources. The term in Part (V) decreases geometrically in the iteration number T, implying that the iterates in Algorithm 1 converge geometrically to a ball of radius determined by the errors from Parts (I)-(IV).

After explaining each part of the upper bound, we now compare it with the convergence rate $\mathcal{O}_{\mathbb{P}}(\frac{p+\log K}{n_k})$ (including $\log K$ here since we consider all the tasks simultaneously) in the single-task learning and reveal how our method performs. With a quick inspection, we can conclude the following:

- The rate of the upper bound is never larger than $\frac{p+\log K}{n_k}$. So, in terms of convergence rate, our method MTL-GMM performs at least as well as single-task learning, regardless of the similarity level h and outlier task fraction ϵ.
- When $n_S \gg n_k$ (large total sample size for tasks in S), p increases with n_k (diverging dimension), $h \ll \sqrt{\frac{p+\log K}{n_k}}$ (sufficient similarity between tasks in S), and $\epsilon \ll \sqrt{(\max_{k=1:K} n_k)/n_k}$ (small fraction of outlier tasks), MTL-GMM attains a faster excess mis-clustering error rate and improves over single-task learning.

The preceding discussions on the upper bounds have demonstrated the superiority of our method. But can we do better? To further evaluate the upper bounds of our method, we next derive complementary minimax lower bounds for both estimation error and excess mis-clustering error. We will show that our method is (nearly) minimax rate optimal in a broad range of regimes.

Theorem 3 (Lower bounds of the estimation error of GMM parameters in multi-task learning). Suppose $\epsilon = \frac{K-s}{K} < 1/3$. Suppose there exists a subset S with $|S| \ge s$ such that $\min_{k\in S} n_k \ge C_1(p + \log K)$ and $\min_{k\in S} \Delta^{(k)} \ge C_2$, where $C_1, C_2 > 0$ are some constants. Then

$$\inf_{\{\widehat{\boldsymbol{\theta}}^{(k)}\}_{k=1}^K} \sup_{S:|S|\ge s} \sup_{\substack{\{\overline{\boldsymbol{\theta}}^{(k)*}\}_{k\in S}\in\overline{\Theta}_S(h) \\ \mathbb{Q}_S}} \mathbb{P}\Bigg(\bigcup_{k\in S}\Bigg\{d(\widehat{\boldsymbol{\theta}}^{(k)}, \boldsymbol{\theta}^{(k)*}) \gtrsim \sqrt{\frac{p}{n_S}} + \sqrt{\frac{\log K}{n_k}}$$

$$+ h \wedge \sqrt{\frac{p+\log K}{n_k} + \frac{\epsilon}{\sqrt{\max_{k=1:K} n_k}}}\Bigg\}\Bigg) \geq \frac{1}{10},$$

$$\inf_{\substack{\{\widehat{\boldsymbol{\mu}}_1^{(k)}, \widehat{\boldsymbol{\mu}}_2^{(k)}\}_{k=1}^K \\ \{\widehat{\boldsymbol{\Sigma}}^{(k)}\}_{k=1}^K}} \sup_{S:|S|\geq s} \sup_{\substack{\{\overline{\boldsymbol{\theta}}^{(k)*}\}_{k\in S}\in\overline{\Theta}_S(h) \\ \mathbb{Q}_S}} \mathbb{P}\Bigg(\bigcup_{k\in S}\Bigg\{\Big(\min_{\pi:[2]\to[2]}\max_{r=1:2}\|\widehat{\boldsymbol{\mu}}_r^{(k)} - \boldsymbol{\mu}_{\pi(r)}^{(k)*}\|_2\Big) \vee \|\widehat{\boldsymbol{\Sigma}}^{(k)} - \boldsymbol{\Sigma}^{(k)*}\|_2$$

$$\gtrsim \sqrt{\frac{p+\log K}{n_k}}\Bigg\}\Bigg) \geq \frac{1}{10}.$$

Theorem 4 (Lower bound of the excess mis-clustering error in multi-task learning). Suppose the same conditions in Theorem 3 hold. Then

$$\inf_{\{\widehat{\mathcal{C}}^{(k)}\}_{k=1}^K} \sup_{S:|S|\geq s} \sup_{\substack{\{\overline{\boldsymbol{\theta}}^{(k)*}\}_{k\in S}\in\overline{\Theta}_S(h) \\ \mathbb{Q}_S}} \mathbb{P}\Bigg(\bigcup_{k\in S}\Bigg\{R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\widehat{\mathcal{C}}^{(k)}) - R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\mathcal{C}_{\overline{\boldsymbol{\theta}}^{(k)*}}) \gtrsim \frac{p}{n_S} + \frac{\log K}{n_k}$$
$$+ h^2 \wedge \frac{p+\log K}{n_k} + \frac{\epsilon^2}{\max_{k=1:K} n_k}\Bigg\}\Bigg) \geq \frac{1}{10}.$$

Comparing the upper and lower bounds in Theorems 1–4, we make several remarks:

- Regarding the estimation of mean vectors $\{\boldsymbol{\mu}_1^{(k)*}, \boldsymbol{\mu}_2^{(k)*}\}_{k\in S}$ and covariance matrices $\{\boldsymbol{\Sigma}^{(k)*}\}_{k\in S}$, the upper and lower bounds match, hence, our method is minimax rate optimal.
- For the estimation error $d(\widehat{\boldsymbol{\theta}}^{(k)}, \boldsymbol{\theta}^{(k)*})$ and excess mis-clustering error $R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\widehat{\mathcal{C}}^{(k)[T]}) - R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\mathcal{C}_{\overline{\boldsymbol{\theta}}^{(k)*}})$ with $k \in S$, the first three terms in the upper and lower bounds match. Only the term involving ϵ in the lower bound differs from that in the upper bound by a factor $\sqrt{p+\log K}$ or $p + \log K$. As a result, in the classical low-dimensional regime where p is bounded, the upper and lower bounds match (up to a logarithmic factor). Therefore, our method is (nearly) minimax rate optimal for estimating $\{\boldsymbol{\theta}^{(k)*}\}_{k\in S}$ and clustering in such a classical regime.
- When the dimension p diverges, there might exist a non-negligible gap between the upper and lower bounds for $d(\widehat{\boldsymbol{\theta}}^{(k)}, \boldsymbol{\theta}^{(k)*})$ and $R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\widehat{\mathcal{C}}^{(k)[T]}) - R_{\overline{\boldsymbol{\theta}}^{(k)*}}(\mathcal{C}_{\overline{\boldsymbol{\theta}}^{(k)*}})$ with $k \in S$. Nevertheless, this only occurs when the fraction of outlier tasks ϵ is above the threshold $\sqrt{\frac{\max_{k=1:K} n_k}{p+\log K}}\big(h^2 \vee \frac{\log K}{n_k} \vee \frac{p}{n_S}\big)$. Below the threshold, our method remains minimax rate optimal even though p is unbounded.
- Does the gap, when it exists, arise from the upper bound or the lower bound? It is the upper bound that sometimes becomes not sharp. As can be seen from the proof of Theorem 1, the term $\epsilon\sqrt{\frac{p+\log K}{\max_{k=1:K} n_k}}$ is due to the estimation of those "center" parameters in Algorithm 1. Recent advances in robust statistics (Chen, Gao, and Ren 2018) have shown that estimators based on statistical depth functions such as Tukey's depth function (Tukey 1975) can achieve optimal minimax rate under Huber's ϵ-contamination model for location and covariance estimation. It might be possible to use depth functions to estimate "center" parameters in our problem and eliminate the factor $\sqrt{p}$ in the upper bound. We leave the rigorous development of optimal robustness as an interesting future research topic. On the other hand, such statistical improvement may come with expensive computation, as depth function-based estimation typically requires solving a challenging non-convex optimization problem.

2.4. Initialization and Cluster Alignment

As specified by Condition (iii) in Assumption 1, our proposed learning procedure requires that for each task in S, initial values of the GMM parameter estimates lie within a distance of SNR-order from the ground truth. This condition can be satisfied by various estimators, such as the method of moments proposed in Ge, Huang, and Kakade (2015) and the robust initialization procedures developed in Jana, Fan, and Kulkarni (2024). In practice, a natural initialization method is to run the standard EM algorithm or other common clustering methods like k-means on each task and use the corresponding estimate as the initial values. We adopted the standard EM algorithm in our numerical experiments, and the numerical results in Section 3 and supplements showed that this practical initialization works quite well. However, in the context of multi-task learning, Condition (iii) further requires a correct alignment of those good initializations from each task, owing to the non-identifiability of GMMs. We discuss in detail the alignment issue in Section 2.4.1 and propose two algorithms to resolve this issue in Section 2.4.2.

2.4.1. The Alignment Issue

Recall that Section 2.1 introduces the binary GMM with parameters $(w^{(k)*}, \boldsymbol{\mu}_1^{(k)*}, \boldsymbol{\mu}_2^{(k)*}, \boldsymbol{\Sigma}^{(k)*})$ for each task $k \in S$. Because the two sets of parameter values $\{(w, \boldsymbol{u}, \boldsymbol{v}, \boldsymbol{\Sigma}), (1-w, \boldsymbol{v}, \boldsymbol{u}, \boldsymbol{\Sigma})\}$ for $(w^{(k)*}, \boldsymbol{\mu}_1^{(k)*}, \boldsymbol{\mu}_2^{(k)*}, \boldsymbol{\Sigma}^{(k)*})$ index the same distribution, a good initialization close to the truth is up to a permutation of the two cluster labels. The permutations in the initialization of different tasks could be different. Therefore, in light of the joint parameter space $\overline{\Theta}_S(h)$ defined in (2) and Condition (iii) in Assumption 1, for given initializations from different tasks, we may need to permute their cluster labels to feed the well-aligned initialization into Algorithm 1.

We further elaborate on the alignment issue using Algorithm 1. The penalization in Step 9 aims to push the estimators $\widehat{\boldsymbol{\beta}}^{(k)[t]}$'s with different k towards each other, which is expected to improve the performance thanks to the similarity among underlying true parameters $\{\boldsymbol{\beta}^{(k)*}\}_{k\in S}$. However, due to the potential permutation of two cluster labels, the vanilla single-task initializations (without alignment) cannot guarantee that the estimators $\{\widehat{\boldsymbol{\beta}}^{(k)[t]}\}_{k\in S}$ at each iteration are all estimating the corresponding $\boldsymbol{\beta}^{(k)*}$'s (some may estimate $-\boldsymbol{\beta}^{(k)*}$'s).

Figure 1 illustrates the alignment issue in the case of two tasks. The left-hand-side situation is ideal where $\widehat{\boldsymbol{\beta}}^{(1)[0]}$, $\widehat{\boldsymbol{\beta}}^{(2)[0]}$ are estimates of $\boldsymbol{\beta}^{(1)*}$, $\boldsymbol{\beta}^{(2)*}$ (which are similar). The right-hand-side situation is problematic because $\widehat{\boldsymbol{\beta}}^{(1)[0]}$, $\widehat{\boldsymbol{\beta}}^{(2)[0]}$ are estimates of $\boldsymbol{\beta}^{(1)*}$, $-\boldsymbol{\beta}^{(2)*}$ (which are not similar). Therefore, after obtaining the initializations from each task, it is necessary

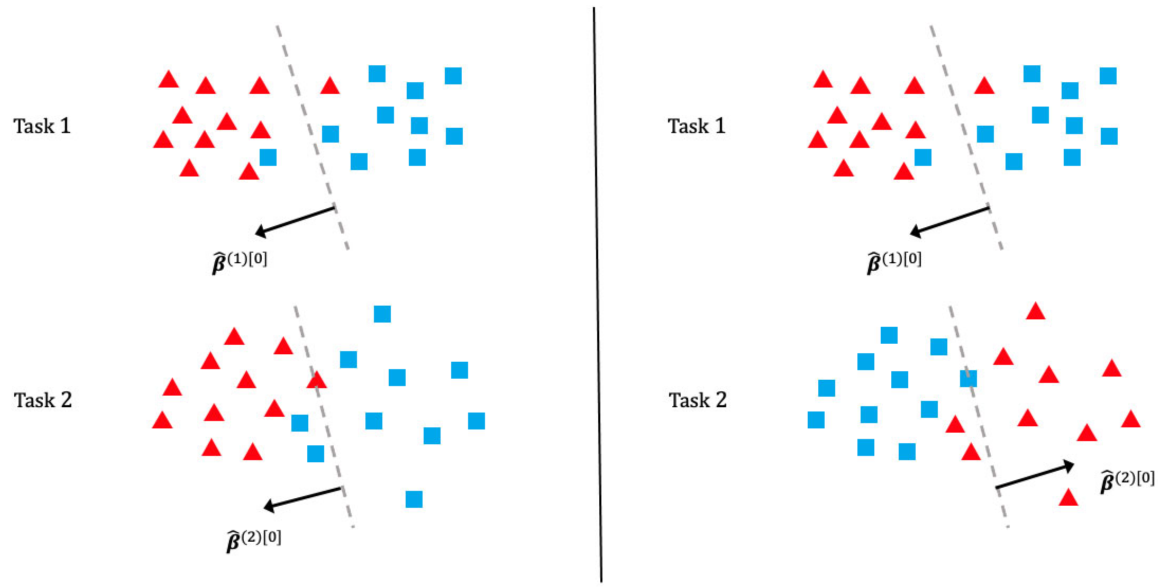

Figure 1. Examples of well-aligned (left) and badly-aligned (right) initializations.

to align their cluster labels to ensure that estimators of similar parameters are correctly put together in the penalization framework in Algorithm 1. We formalize the problem and provide two solutions in the next subsection.

2.4.2. Two Alignment Algorithms

Suppose $\{\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k=1}^K$ are the initial estimates of discriminant coefficients with potentially bad alignment for $k \in S$. Note that a good initialization and alignment is not required (in fact, it is not even well defined) for the outlier tasks in S^c, because they can be from arbitrary distributions. However, since S is unknown, we will have to address the alignment issue for tasks in S based on initial estimates from all the tasks. For binary GMMs, each alignment of $\{\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k=1}^K$ can be represented by a K-dimensional Rademacher vector $\boldsymbol{r} \in \{\pm 1\}^K$. Define the ideal alignment as $r_k^* = \arg\min_{r_k=\pm 1} \|r_k\widehat{\boldsymbol{\beta}}^{(k)[0]} - \boldsymbol{\beta}^{(k)*}\|_2, k \in S$. The goal is to recover the well-aligned initializers $\{r_k^*\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k\in S}$ from the initial estimates $\{\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k=1}^K$ (equivalently, to recover $\{r_k^*\}_{k\in S}$), which can then be fed into Algorithm 1. Once $\{\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k\in S}$ are well aligned, other initial estimates in Algorithm 1 will be automatically well aligned.

In the following, we will introduce two alignment algorithms. The first one is the "exhaustive search" method (Algorithm 2), where we search among all possible alignments to find the best one. The second one is the "greedy search" method (Algorithm 3 in Section S.1 of the supplements), where we flip the sign of $\widehat{\boldsymbol{\beta}}^{(k)[0]}$ in a greedy way to recover $\{r_k^*\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k\in S}$. Both methods are proved to recover $\{r_k^*\}_{k\in S}$ under mild conditions. The conditions required by the "exhaustive search" method are slightly weaker than those required by the "greedy search" method. As for computational complexity, the latter enjoys a linear time complexity $\mathcal{O}(K)$, while the former suffers from an exponential time complexity $\mathcal{O}(2^K)$ due to optimization over all possible 2^K alignments. Due to space constraints, we will introduce only

Algorithm 2: Exhaustive search for the alignment

Input: Initialization $\{\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k=1}^K$
$\widehat{\boldsymbol{r}} \leftarrow \arg\min_{\boldsymbol{r}\in\{\pm 1\}^K} \mathrm{score}(\boldsymbol{r})$
Output: $\widehat{\boldsymbol{r}}$

the "exhaustive search" method, while the details of the "greedy search" algorithm are provided in Section S.1 of the supplementary materials.

To this end, for a given alignment $\boldsymbol{r} = \{r_k\}_{k=1}^K \in \{\pm 1\}^K$ with the correspondingly aligned estimates $\{r_k\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k=1}^K$, define its alignment score as

$$\mathrm{score}(\boldsymbol{r}) = \sum_{1\le k_1\neq k_2\le K} \|r_{k_1}\widehat{\boldsymbol{\beta}}^{(k_1)[0]} - r_{k_2}\widehat{\boldsymbol{\beta}}^{(k_2)[0]}\|_2.$$

The intuition is that as long as the initializations $\{\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k\in S}$ are close to the ground truth, a smaller score indicates less difference among $\{r_k\widehat{\boldsymbol{\beta}}^{(k)[0]}\}_{k\in S}$, which implies a better alignment. The score can be used to evaluate the quality of an alignment. Note that the score is defined in a symmetric way, that is, $\mathrm{score}(\boldsymbol{r}) = \mathrm{score}(-\boldsymbol{r})$. The exhaustive search algorithm is presented in Algorithm 2, where scores of all alignments are calculated, and the alignment that minimizes the score is output. Since the score is symmetric, there are at least two alignments with the minimum score. The algorithm can arbitrarily choose and output one of them.

The following theorem reveals that the exhaustive search algorithm can successfully find the ideal alignment under mild conditions.

Theorem 5 (Alignment correctness for Algorithm 2). Assume that

(i) $\epsilon < \frac{1}{3}$;

(ii) $\min_{k\in S} \|\boldsymbol{\beta}^{(k)*}\|_2 \geq \frac{4(1-\epsilon)}{1-3\epsilon}h + \frac{2(2-\epsilon)}{1-3\epsilon}\max_{k\in S}\big(\|\widehat{\boldsymbol{\beta}}^{(k)[0]} - \boldsymbol{\beta}^{(k)*}\|_2 \wedge \|\widehat{\boldsymbol{\beta}}^{(k)[0]} + \boldsymbol{\beta}^{(k)*}\|_2\big)$, where $\epsilon = \frac{K-|S|}{K}$ is the outlier task proportion introduced in Theorem 1, and h is the similarity level of discriminant coefficient in (2). Then the output of Algorithm 2 satisfies

$$\widehat{r}_k = r_k^* \text{ for all } k \in S \quad \text{or} \quad \widehat{r}_k = -r_k^* \text{ for all } k \in S$$

Remark 2. The conditions imposed in Theorem 5 are *no stronger* than conditions required by Theorem 1. First of all, Condition (i) is also required in Theorem 1. Moreover, from the definition of h in (2), it is bounded by a constant. This, together with Conditions (iii) and (iv) in Assumption 1, implies Condition (ii) in Theorem 5.

Remark 3. With Theorem 5, we can relax the original Condition (iii) in Assumption 1 to the following condition:

For all $k \in S$, either of the following two conditions holds with a sufficiently small constant C_3:

(a) $\|\widehat{\boldsymbol{\beta}}^{(k)[0]} - \boldsymbol{\beta}^{(k)*}\|_2 \vee \|\widehat{\boldsymbol{\mu}}_1^{(k)[0]} - \boldsymbol{\mu}_1^{(k)*}\|_2 \vee \|\widehat{\boldsymbol{\mu}}_2^{(k)[0]} - \boldsymbol{\mu}_2^{(k)*}\|_2 \leq C_3 \min_{k\in S} \Delta^{(k)}$, $|\widehat{w}^{(k)[0]} - w^{(k)*}| \leq c_w/2$;

(b) $\|\widehat{\boldsymbol{\beta}}^{(k)[0]} + \boldsymbol{\beta}^{(k)*}\|_2 \vee \|\widehat{\boldsymbol{\mu}}_1^{(k)[0]} - \boldsymbol{\mu}_2^{(k)*}\|_2 \vee \|\widehat{\boldsymbol{\mu}}_2^{(k)[0]} - \boldsymbol{\mu}_1^{(k)*}\|_2 \leq C_3 \min_{k\in S} \Delta^{(k)}$, $|1 - \widehat{w}^{(k)[0]} - w^{(k)*}| \leq c_w/2$.

In the relaxed version, the initialization for each task only needs to be good up to an *arbitrary* permutation, while in the original version, the initialization for each task needs to be good under *the same permutation*.

In contrast with supervised MTL, the alignment issue commonly exists in unsupervised MTL. It generally occurs when aggregating information (up to latent label permutation) across different tasks. Alignment pre-processing is thus necessary and important. However, to our knowledge, there is no formal discussion regarding alignment in the existing literature of unsupervised MTL (Gu, Li, and Han 2011; Zhang and Zhang 2011; Yang et al. 2014; Zhang et al. 2018; Dieuleveut et al. 2021; Marfoq et al. 2021). Our treatment of alignment is an important step forward in this field. Our algorithms can potentially be extended to other unsupervised MTL scenarios, and we will leave it for future studies.

Lastly, we emphasize that our alignment algorithms, including Algorithm 2 and the greedy search algorithm described in Section S.1, do not require any specific initialization procedures. For instance, as Condition (ii) in Theorem 5 shows, Algorithm 2 performs well given a reasonably good initialization, and better initializations allow for weaker assumptions on the SNR. As discussed at the beginning of Section 2.4, several estimators in the literature satisfy this requirement. In practice, we prefer simple choices such as those obtained from k-means or the standard single-task EM algorithm.

3. Numerical Experiments

In Section 3.1, we present a simulation study for binary GMMs, followed by a real-data analysis on the Human Activity Recognition dataset for a binary clustering problem in Section 3.2. Due to space limitations, additional simulation studies in various settings, experiments on tuning parameter selection, a multi-cluster analysis of the Human Activity Recognition dataset, and a real-data study on handwritten digits clustering are provided in Section S.5 of the supplementary materials.

As mentioned earlier, we also explored the transfer learning versions of our algorithms in the supplementary material. Given their close similarity in framework to the multi-task versions, we expect them to exhibit comparable performance and therefore do not include them here for comparison.

3.1. Simulations

In this section, we present a simulation study of our multi-task learning procedure MTL-GMM, that is, Algorithm 1. The tuning parameter $\kappa \in (0, 1)$ is set as $1/3$, and the value of C_λ is determined by a 10-fold cross-validation based on the log-likelihood of the final fitted model. The candidates of C_λ are chosen in a data-driven way, which is described in detail in Section S.5.1.7 of the supplements. All the experiments in this section are implemented in R. Function `Mclust` in R package `mclust` is called to fit a single GMM. We also conducted two additional simulation studies. Due to space constraints, we included these in Section S.5 of the supplementary materials.

We consider a binary GMM setting. There are $K = 10$ tasks, each of which has a sample size of $n_k = 100$ and a dimension of $p = 15$. When $k \in S$, we generate each $w^{(k)*}$ from Unif(0.1, 0.9) and $\boldsymbol{\mu}_1^{(k)*}$ from $(2, 2, \mathbf{0}_{p-2})^\top + h/2 \cdot (\boldsymbol{\Sigma}^{(k)*})^{-1}\boldsymbol{u}$, where $\boldsymbol{u} \sim$ Unif$(\{\boldsymbol{u} \in \mathbb{R}^p : \|\boldsymbol{u}\|_2 = 1\})$, $\boldsymbol{\Sigma}^{(k)*} = (0.2^{|i-j|})_{p\times p}$, and let $\boldsymbol{\mu}_2^{(k)*} = -\boldsymbol{\mu}_1^{(k)*}$. When $k \notin S$, the distributions still follow GMM, but we generate each $w^{(k)*}$ from Unif(0.2, 0.4) and $\boldsymbol{\mu}_1^{(k)*}$ from Unif$(\{\boldsymbol{u} \in \mathbb{R}^p : \|\boldsymbol{u}\|_2 = 5\})$, and let $\boldsymbol{\mu}_2^{(k)*} = -\boldsymbol{\mu}_1^{(k)*}$, $\boldsymbol{\Sigma}^{(k)*} = (0.5^{|i-j|})_{p\times p}$. In this setup, it is clear that h quantifies the similarity among tasks in S, and tasks in S^c have very distinct distributions and can be viewed as outlier tasks. For a given $\epsilon \in [0, 1)$, the outlier task index set S^c in each replication is uniformly sampled from all subsets of $1 : K$ with cardinality $K\epsilon$. We consider two cases:

(i) No outlier tasks ($\epsilon = 0$), and h changes from 0 to 10 with increment 1;
(ii) 2 outlier tasks ($\epsilon = 0.2$), and h changes from 0 to 10 with increment 1.

We fit Single-task-GMM on each separate task, Pooled-GMM on the merged data of all tasks, and our MTL-GMM in Algorithm 1 coupled with the exhaustive search for the alignment in Algorithm 2. The performances of all three methods are evaluated by the estimation error of $w^{(k)*}$, $\boldsymbol{\mu}_1^{(k)*}$, $\boldsymbol{\mu}_2^{(k)*}$, $\boldsymbol{\beta}^{(k)*}$, $\delta^{(k)*}$, and $\boldsymbol{\Sigma}^{(k)*}$, as well as the empirical mis-clustering error calculated on a test dataset of size 500, for tasks in S. Due to page limit, we only present the estimation error of $\boldsymbol{\beta}^{(k)*}$ and the mis-clustering error here, and leave the others to Section S.5.1.1 of the supplements. These two errors are the maximum errors over tasks in S. For each setting, the simulation is replicated 200 times, and the average of the maximum errors and the standard deviation are reported in Figure 2.

When there are no outlier tasks, it can be seen that MTL-GMM and Pooled-GMM are competitive when h is small (i.e., the tasks are similar), and they outperform Single-task-GMM. As h increases (i.e., tasks become more heterogeneous), MTL-

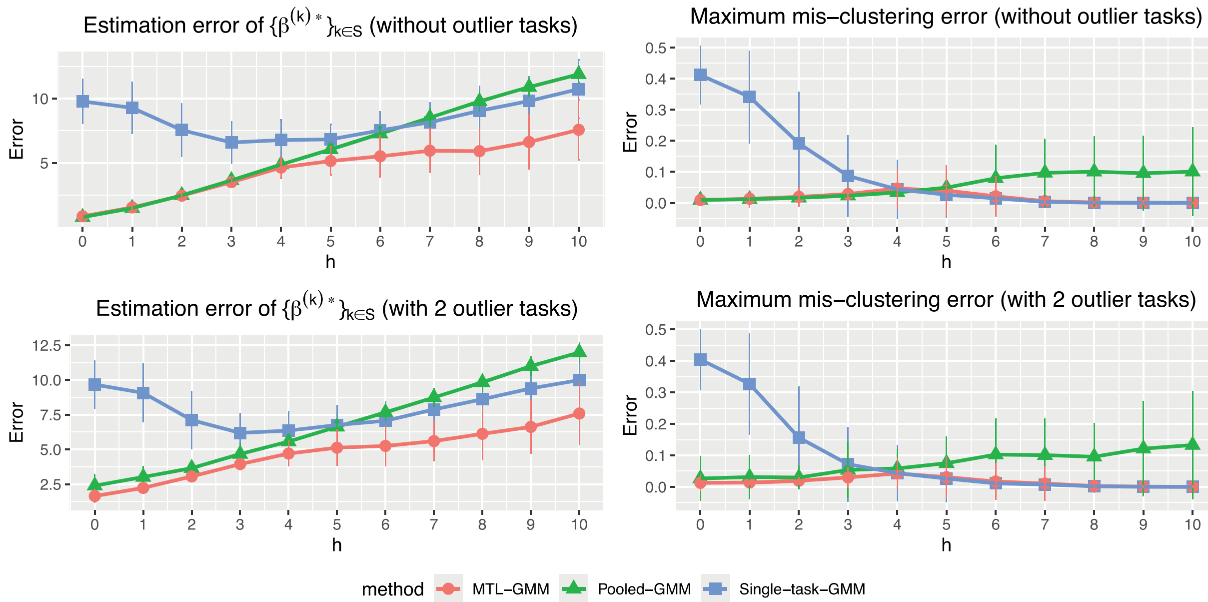

Figure 2. The performance of different methods in Simulation 1 under different outlier proportions. The upper panel shows the performance without outlier tasks ($\epsilon = 0$), and the lower panel shows the performance with two outlier tasks ($\epsilon = 0.2$). h changes from 0 to 10 with increment 1. Estimation error of $\{\boldsymbol{\beta}^{(k)*}\}_{k\in S}$ stands for $\max_{k\in S}(\|\widehat{\boldsymbol{\beta}}^{(k)[T]} - \boldsymbol{\beta}^{(k)*}\|_2 \wedge \|\widehat{\boldsymbol{\beta}}^{(k)[T]} + \boldsymbol{\beta}^{(k)*}\|_2)$ and maximum mis-clustering error represents the maximum empirical mis-clustering error rate calculated on the test set of tasks in S.

GMM starts to outperform Pooled-GMM by a large margin. Moreover, MTL-GMM is significantly better than Single-task-GMM in terms of both estimation and mis-clustering errors over a wide range of h. These comparisons demonstrate that MTL-GMM not only effectively uses the unknown similarity structure among tasks, but also adapts to it. When the outlier tasks exist, even when h is very small, MTL-GMM still performs better than Pooled-GMM, showing the robustness of MTL-GMM against a fraction of outlier tasks.

3.2. A Real-Data Study

Human Activity Recognition (HAR) Using Smartphones Dataset contains the data collected from 30 volunteers when they performed six activities (walking, walking upstairs, walking downstairs, sitting, standing, and laying) wearing a smartphone (Anguita et al. 2013). Each observation has 561 time and frequency domain variables. Each volunteer can be viewed as a task, and the sample size of each task varies from 281 to 409. The original dataset is available at UCI Machine Learning Repository: *https://archive.ics.uci.edu/ml/datasets/human+activity+recognition+using+smartphones*.

Here, we first focus on two activities, standing and laying, and perform clustering without the label information, to test our method in the binary case. This is a binary MTL clustering problem with 30 tasks. The sample size of each task varies from 95 to 179. For each task, in each replication, we use 90% of the samples as training data and hold 10% of the samples as test data.

We first run a principal component analysis (PCA) on the training data of each task and project both the training and test data onto the first 15 principal components. PCA has often been used for dimension reduction in pre-processing the HAR dataset (Walse, Dharaskar, and Thakare 2016; Aljarrah and Ali 2019; Duan and Wang 2023). We fit Single-task-GMM on each task separately, Pooled-GMM on merged data from 30 tasks, and our MTL-GMM with the greedy label swapping alignment algorithm. The performance of the three methods is evaluated by the mis-clustering error rate on the test data of all 30 tasks. The maximum and average mis-clustering errors among the 30 tasks are calculated in each replication. The mean and standard deviation of these two errors over 200 replications are reported in Table 1. To better display the clustering performance on each task, we further generate the boxplot of mis-clustering errors of 30 tasks (averaged over 200 replications) for each method in Figure 3. It is clear that MTL-GMM outperforms both Pooled-GMM and Single-task-GMM. Note that MTL-GMM requires only the discriminant coefficients to be similar across tasks in order to improve clustering performance over Single-task-GMM, whereas Pooled-GMM relies on stronger assumptions—namely, similarity in both the mean vectors and the covariance matrices. In the current context, individual differences in height, weight, and body shape can affect the observed data patterns, even when the same gestures are performed. This practical heterogeneity makes the assumptions of Pooled-GMM less realistic and, consequently, favors the use of MTL-GMM in this real-data study.

Table 1. Maximum and average mis-clustering errors and standard deviations (numbers in the parentheses) in the HAR dataset.

Method	Single-task	Pooled	MTL
Maximum error	0.49 (0.02)	0.38 (0.12)	0.36 (0.09)
Average error	0.28 (0.02)	0.15 (0.17)	0.03 (0.01)

Due to space constraints, additional results from this real-data study focusing on the multi-component GMM extension and an additional application to handwritten digit recognition are provided in Section S.5.2 of the supplementary materials. We refer interested readers to that section for further details.

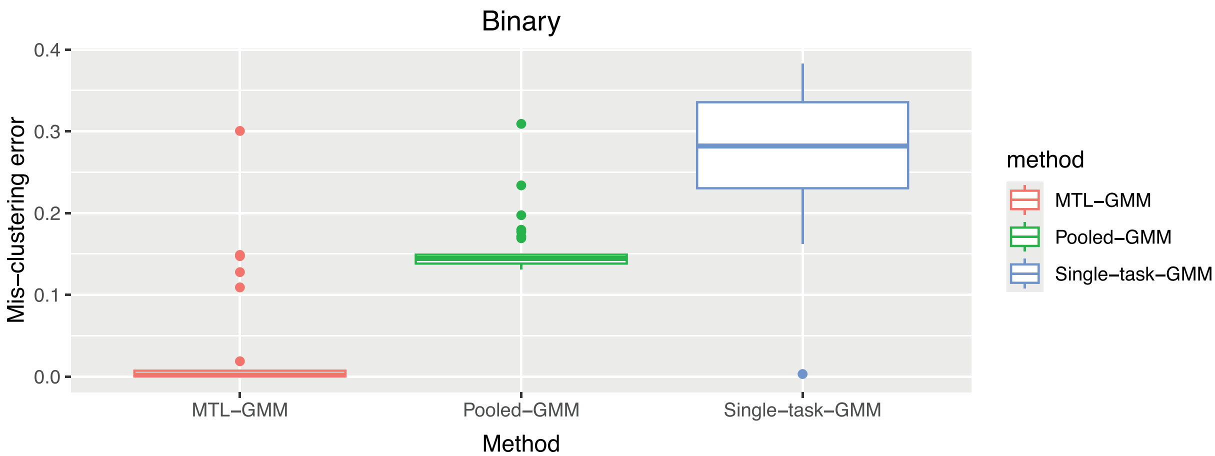

Figure 3. Boxplots of mis-clustering errors of 30 tasks for each method for HAR dataset.

4. Discussions

We briefly comment on several extensions that are developed in the supplementary materials. First, Section S.2 generalizes the algorithms and theory from binary to multi-component GMMs with $R \geq 3$ clusters. Section S.3.1 considers settings where tasks have different numbers of clusters; we discuss both a direct extension that aggregates only over clusters shared across tasks and a representation-learning approach in which task-specific component means lie in a common low-dimensional subspace. Section S.3.2 discusses how to select the number of clusters when it is unknown and the impact of mis-specification. Section S.3.3 discusses the possibility of relaxing the common-covariance assumption within each task, leading to quadratic decision boundaries and suggesting that cross-task similarity should be imposed jointly on the linear and quadratic terms of the classifier. Finally, Section S.4 develops the transfer learning counterpart of our method, including the algorithms, theoretical guarantees, and label-alignment procedures, where information from multiple source tasks is robustly aggregated to improve performance on a target task. We refer interested readers to these sections for further details.

Supplementary Materials

The supplementary materials present a greedy label-alignment algorithm; extensions of the framework to multi-cluster GMMs, varying cluster numbers, and heterogeneous covariances; a complete development of the transfer-learning counterpart; and additional simulation and real-data results. They also contain full proofs of all theorems stated in the main paper and the supplement, together with the supporting general lemmas.

Acknowledgments

We thank the co-Editor Dylan Small, the AE, and the anonymous referees for their insightful comments, which greatly improved the scope and quality of the article. All experiments were conducted on Ginsburg HPC Cluster of Columbia University.

Disclosure Statement

The authors report there are no competing interests to declare.

Funding

Haolei Weng was partially supported by NSF grant DMS-2210505. Lucy Xia was partially supported by RGC GRF grant 16308625. Yang Feng was partially supported by NSF Grant DMS-2324489 and NIH Grant 1R21AG074205-01.

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References

Aljarrah, A. A., and Ali, A. H. (2019), "Human Activity Recognition Using PCA and Bilstm Recurrent Neural Networks," in *2019 2nd International Conference on Engineering Technology and its Applications (IICETA)*, pp. 156–160, IEEE. [11]

Anderson, T. W. (1958), *An Introduction to Multivariate Statistical Analysis: Wiley Series in Probability and Mathematical Statistics: Probability and Mathematical Statistics*, New York, NY: Wiley. [1,6]

Ando, R. K., Zhang, T., and Bartlett, P. (2005), "A Framework for Learning Predictive Structures from Multiple Tasks and Unlabeled Data," *Journal of Machine Learning Research*, 6, 1817–1853. [2]

Anguita, D., Ghio, A., Oneto, L., Parra Perez, X., and Reyes Ortiz, J. L. (2013), "A Public Domain Dataset for Human Activity Recognition Using Smartphones," in *Proceedings of the 21st International European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning*, pp. 437–442. [11]

Argyriou, A., Evgeniou, T., and Pontil, M. (2008), "Convex Multi-Task Feature Learning," *Machine Learning*, 73, 243–272. [2]

Azizyan, M., Singh, A., and Wasserman, L. (2013), "Minimax Theory for High-Dimensional Gaussian Mixtures with Sparse Mean Separation," in *Advances in Neural Information Processing Systems*. [6]

Balakrishnan, S., Wainwright, M. J., and Yu, B. (2017), "Statistical Guarantees for the EM Algorithm: From Population to Sample-based Analysis," *The Annals of Statistics*, 45, 77–120. [2,6]

Bastani, H. (2021), "Predicting with Proxies: Transfer Learning in High Dimension," *Management Science*, 67, 2964–2984. [2,4]

Baum, L. E., Petrie, T., Soules, G., and Weiss, N. (1970), "A Maximization Technique Occurring in the Statistical Analysis of Probabilistic Functions of Markov Chains," *The Annals of Mathematical Statistics*, 41, 164–171. [2]

Cai, T., and Liu, W. (2011), "A Direct Estimation Approach to Sparse Linear Discriminant Analysis," *Journal of the American Statistical Association*, 106, 1566–1577. [4]

Cai, T. T., Ma, J., and Zhang, L. (2019), "Chime: Clustering of High-Dimensional Gaussian Mixtures with EM Algorithm and its Optimality," *The Annals of Statistics*, 47, 1234–1267. [2,3,4,6,7]

Carroll, R. J., Fan, J., Gijbels, I., and Wand, M. P. (1997), "Generalized Partially Linear Single-Index Models," *Journal of the American Statistical Association*, 92, 477–489. [6]

Chattopadhyay, R., Sun, Q., Fan, W., Davidson, I., Panchanathan, S., and Ye, J. (2012), "Multisource Domain Adaptation and its Application to Early Detection of Fatigue," *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 6, 1–26. [2]

Chen, E. Y., Jordan, M. I., and Li, S. (2022), "Transferred q-Learning," arXiv preprint arXiv:2202.04709. [2]

Chen, M., Gao, C., and Ren, Z. (2018), "Robust Covariance and Scatter Matrix Estimation under Huber's Contamination Model," *The Annals of Statistics*, 46, 1932–1960. [8]

Dai, W., Yang, Q., Xue, G., and Yu, Y. (2007), "Boosting for Transfer Learning," in *ACM International Conference Proceeding Series* (Vol. 227), pp. 193. [2]

Dai, W., Yang, Q., Xue, G.-R., and Yu, Y. (2008), "Self-Taught Clustering," in *Proceedings of the 25th International Conference on Machine Learning*, pp. 200–207. [2]

Dasgupta, S., and Schulman, L. J. (2000), "A Two-Round Variant of EM for Gaussian Mixtures," in *Proceedings of the Sixteenth Conference on Uncertainty in Artificial Intelligence*, pp. 152–159. [2,6]

Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977), "Maximum Likelihood from Incomplete Data via the EM Algorithm," *Journal of the Royal Statistical Society*, Series B, 39, 1–22. [2]

Dieuleveut, A., Fort, G., Moulines, E., and Robin, G. (2021), "Federated-EM with Heterogeneity Mitigation and Variance Reduction," in *Advances in Neural Information Processing Systems* (Vol. 34), pp. 29553–29566. [3,10]

Donoho, D., and Montanari, A. (2016), "High Dimensional Robust m-estimation: Asymptotic Variance via Approximate Message Passing," *Probability Theory and Related Fields*, 166, 935–969. [5]

Duan, Y., and Wang, K. (2023), "Adaptive and Robust Multi-Task Learning," *The Annals of Statistics*, 51, 2015–2039. [2,4,5,6,11]

Efron, B. (1975), "The Efficiency of Logistic Regression Compared to Normal Discriminant Analysis," *Journal of the American Statistical Association*, 70, 892–898. [1]

Evgeniou, T., and Pontil, M. (2004), "Regularized Multi–Task Learning," in *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 109–117. [2]

Fan, J., Feng, Y., and Tong, X. (2012), "A Road to Classification in High Dimensional Space: The Regularized Optimal Affine Discriminant," *Journal of the Royal Statistical Society*, Series B, 74, 745–771. [4]

Forgy, E. W. (1965), "Cluster Analysis of Multivariate Data: Efficiency versus Interpretability of Classifications," *Biometrics*, 21, 768–769. [1]

Ge, R., Huang, Q., and Kakade, S. M. (2015), "Learning Mixtures of Gaussians in High Dimensions," in *Proceedings of the Forty-Seventh Annual ACM Symposium on Theory of Computing*, pp. 761–770. [2,8]

Gu, Q., Li, Z., and Han, J. (2011), "Learning a Kernel for Multi-Task Clustering," in *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 25), pp. 368–373. [2,3,10]

Hartley, H. O. (1958), "Maximum Likelihood Estimation from Incomplete Data," *Biometrics*, 14, 174–194. [2]

Hasselblad, V. (1966), "Estimation of Parameters for a Mixture of Normal Distributions," *Technometrics*, 8, 431–444. [2]

Hastie, T., Tibshirani, R., and Friedman, J. H. (2009), *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (Vol. 2), New York: Springer. [1]

Hsu, D., and Kakade, S. M. (2013), "Learning Mixtures of Spherical Gaussians: Moment Methods and Spectral Decompositions," in *Proceedings of the 4th Conference on Innovations in Theoretical Computer Science*, pp. 11–20. [2]

Huber, P. J. (1964), "Robust Estimation of a Location Parameter," *The Annals of Mathematical Statistics*, 35, 73–101. [4]

Jain, A. K., and Dubes, R. C. (1988), *Algorithms for Clustering Data*, Englewood Cliffs, NJ: Prentice-Hall, Inc. [1]

Jalali, A., Sanghavi, S., Ruan, C., and Ravikumar, P. (2010), "A Dirty Model for Multi-Task Learning," in *Advances in Neural Information Processing Systems* (Vol. 23). [2]

Jana, S., Fan, J., and Kulkarni, S. (2024), "A Provable Initialization and Robust Clustering Method for General Mixture Models," arXiv preprint arXiv:2401.05574. [8]

Jin, J., Ke, Z. T., and Wang, W. (2017), "Phase Transitions for High Dimensional Clustering and Related Problems," *The Annals of Statistics*, 45, 2151–2189. [2]

Kalai, A. T., Moitra, A., and Valiant, G. (2010), "Efficiently Learning Mixtures of Two Gaussians," in *Proceedings of the Forty-Second ACM Symposium on Theory of Computing*, pp. 553–562. [2]

Konstantinov, N., Frantar, E., Alistarh, D., and Lampert, C. (2020), "On the Sample Complexity of Adversarial Multi-Source PAC Learning," in *International Conference on Machine Learning*, pp. 5416–5425, PMLR. [4]

Kwon, J., and Caramanis, C. (2020), "The EM Algorithm Gives Sample-Optimality for Learning Mixtures of Well-Separated Gaussians," in *Conference on Learning Theory*, pp. 2425–2487. [2,4]

Lawrence, N. D., and Platt, J. C. (2004), "Learning to Learn with the Informative Vector Machine," in *Proceedings of the Twenty-First International Conference on Machine Learning*, p. 65. [2]

Lee, K., Guillemot, L., Yue, Y., Kramer, M., and Champion, D. (2012), "Application of the Gaussian Mixture Model in Pulsar Astronomy-Pulsar Classification and Candidates Ranking for the Fermi 2fgl Catalogue," *Monthly Notices of the Royal Astronomical Society*, 424, 2832–2840. [1]

Li, R., and Liang, H. (2008), "Variable Selection in Semiparametric Regression Modeling," *The Annals of Statistics*, 36, 261–286. [6]

Li, S., Cai, T., and Duan, R. (2023), "Targeting Underrepresented Populations in Precision Medicine: A Federated Transfer Learning Approach," *Annals of Applied Statistics*, 17, 2970–2992. [2,4]

Li, S., Cai, T. T., and Li, H. (2021), "Transfer Learning for High-Dimensional Linear Regression: Prediction, Estimation and Minimax Optimality," *Journal of the Royal Statistical Society*, Series B, 84, 149–173. [4]

——— (2022a), "Estimation and Inference with Proxy Data and its Genetic Applications," arXiv preprint arXiv:2201.03727. [2]

——— (2022b), "Transfer Learning in Large-Scale Gaussian Graphical Models with False Discovery Rate Control," *Journal of the American Statistical Association*, 118, 2171–2183. [2]

Li, W., Duan, L., Xu, D., and Tsang, I. W. (2013), "Learning with Augmented Features for Supervised and Semi-Supervised Heterogeneous Domain Adaptation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36, 1134–1148. [2]

Lin, H., and Reimherr, M. (2022), "On Transfer Learning in Functional Linear Regression," arXiv preprint arXiv:2206.04277. [2,4]

Mai, Q., Yang, Y., and Zou, H. (2019), "Multiclass Sparse Discriminant Analysis," *Statistica Sinica*, 29, 97–111. [4]

Mai, Q., Zou, H., and Yuan, M. (2012), "A Direct Approach to Sparse Discriminant Analysis in Ultra-High Dimensions," *Biometrika*, 99, 29–42. [4]

Marfoq, O., Neglia, G., Bellet, A., Kameni, L., and Vidal, R. (2021), "Federated Multi-Task Learning Under a Mixture of Distributions," in *Advances in Neural Information Processing Systems* (Vol. 34), pp. 15434–15447. [2,3,10]

McLachlan, G. J., and Krishnan, T. (2007), *The EM Algorithm and Extensions*, New York: Wiley. [2]

Meng, X.-L., and Rubin, D. B. (1994), "On the Global and Componentwise Rates of Convergence of the EM Algorithm," *Linear Algebra and its Applications*, 199, 413–425. [2]

Mihalkova, L., Huynh, T., and Mooney, R. J. (2007), "Mapping and Revising Markov Logic Networks for Transfer Learning," in *Proceedings of the 22nd National Conference on Artificial Intelligence* (Vol. 1), pp. 608–614. [2]

Murtagh, F., and Contreras, P. (2012), "Algorithms for Hierarchical Clustering: An Overview," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 2, 86–97. [1]

Ng, A., Jordan, M., and Weiss, Y. (2001), "On Spectral Clustering: Analysis and an Algorithm," in *Advances in Neural Information Processing Systems* (Vol. 14). [1]

Obozinski, G., Taskar, B., and Jordan, M. (2006), "Multi-Task Feature Selection," Statistics Department, UC Berkeley, Technical Report, 2, 2. [2]

Pan, S. J., and Yang, Q. (2009), "A Survey on Transfer Learning," *IEEE Transactions on Knowledge and Data Engineering*, 22, 1345–1359. [2]

Pearson, K. (1894), "Contributions to the Mathematical Theory of Evolution," *Philosophical Transactions of the Royal Society of London. A*, 185, 71–110. [2]

Redner, R. A., and Walker, H. F. (1984), "Mixture Densities, Maximum Likelihood and the EM Algorithm," *SIAM Review*, 26, 195–239. [2,4,6]

Scott, A. J., and Symons, M. J. (1971), "Clustering Methods based on Likelihood Ratio Criteria," *Biometrics*, 27, 387–397. [1]

She, Y., and Owen, A. B. (2011), "Outlier Detection Using Nonconvex Penalized Regression," *Journal of the American Statistical Association*, 106, 626–639. [5]

Sundberg, R. (1974), "Maximum Likelihood Theory for Incomplete Data from An Exponential Family," *Scandinavian Journal of Statistics*, 1, 49–58. [2]

Thrun, S., and O'Sullivan, J. (1996), "Discovering Structure in Multiple Learning Tasks: The TC Algorithm," in *ICML* (Vol. 96), pp. 489–497. [2]

Tian, Y., and Feng, Y. (2023), "Transfer Learning Under High-Dimensional Generalized Linear Models," *Journal of the American Statistical Association*, 118, 2684–2697. [2,4]

Tukey, J. W. (1975), "Mathematics and the Picturing of Data," in *Proceedings of the International Congress of Mathematicians, Vancouver, 1975* (Vol. 2), pp. 523–531. [8]

Vempala, S., and Wang, G. (2004), "A Spectral Algorithm for Learning Mixture Models," *Journal of Computer and System Sciences*, 68, 841–860. [2]

Walse, K. H., Dharaskar, R. V., and Thakare, V. M. (2016), "PCA based Optimal ANN Classifiers for Human Activity Recognition Using Mobile Sensors Data," in *Proceedings of First International Conference on Information and Communication Technology for Intelligent Systems: Volume 1*, pp. 429–436, Springer. [11]

Wang, R., Zhou, J., Jiang, H., Han, S., Wang, L., Wang, D., and Chen, Y. (2021), "A General Transfer Learning-based Gaussian Mixture Model for Clustering," *International Journal of Fuzzy Systems*, 23, 776–793. [2]

Wang, Z., Gu, Q., Ning, Y., and Liu, H. (2014), "High Dimensional Expectation-Maximization Algorithm: Statistical Optimization and Asymptotic Normality," arXiv preprint arXiv:1412.8729. [2]

Wang, Z., Song, Y., and Zhang, C. (2008), "Transferred Dimensionality Reduction," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pp. 550–565, Springer. [2]

Witten, D. M., and Tibshirani, R. (2011), "Penalized Classification Using Fisher's Linear Discriminant," *Journal of the Royal Statistical Society*, Series B, 73, 753–772. [4]

Wu, C. J. (1983), "On the Convergence Properties of the EM Algorithm," *The Annals of statistics*, 11, 95–103. [2]

Xu, J., Hsu, D. J., and Maleki, A. (2016), "Global Analysis of Expectation Maximization for Mixtures of Two Gaussians," in *Advances in Neural Information Processing Systems* (Vol. 29). [2]

Xu, K., and Bastani, H. (2021), "Learning Across Bandits in High Dimension via Robust Statistics," arXiv preprint arXiv:2112.14233. [2,4]

Yan, B., Yin, M., and Sarkar, P. (2017), "Convergence of Gradient EM on Multi-Component Mixture of Gaussians," in *Advances in Neural Information Processing Systems* (Vol. 30). [2]

Yang, M.-H., and Ahuja, N. (1998), "Gaussian Mixture Model for Human Skin Color and its Applications in Image and Video Databases," in *Storage and Retrieval for Image and Video Databases VII*, (Vol. 3656), pp. 458–466, SPIE. [1]

Yang, Y., Ma, Z., Yang, Y., Nie, F., and Shen, H. T. (2014), "Multitask Spectral Clustering by Exploring Intertask Correlation," *IEEE Transactions on Cybernetics*, 45, 1083–1094. [2,3,10]

Zhang, J., and Zhang, C. (2011), "Multitask Bregman Clustering," *Neurocomputing*, 74, 1720–1734. [2,3,4,10]

Zhang, Q., and Chen, J. (2022), "Distributed Learning of Finite Gaussian Mixtures," *Journal of Machine Learning Research*, 23, 1–40. [2]

Zhang, X., Blanchet, J., Ghosh, S., and Squillante, M. S. (2022), "A Class of Geometric Structures in Transfer Learning: Minimax Bounds and Optimality," in *International Conference on Artificial Intelligence and Statistics*, pp. 3794–3820, PMLR. [2]

Zhang, X., Zhang, X., and Liu, H. (2015), "Smart Multitask Bregman Clustering and Multitask Kernel Clustering," *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 10, 1–29. [4]

Zhang, X., Zhang, X., Liu, H., and Luo, J. (2018), "Multi-Task Clustering with Model Relation Learning," in *Proceedings of the 27th International Joint Conference on Artificial Intelligence*, pp. 3132–3140. [2,3,10]

Zhang, Y., and Yang, Q. (2021), "A Survey on Multi-Task Learning," *IEEE Transactions on Knowledge and Data Engineering*, 34, 5586–5609. [2]

Zhao, R., Li, Y., and Sun, Y. (2020), "Statistical Convergence of the EM Algorithm on Gaussian Mixture Models," *Electronic Journal of Statistics*, 14, 632–660. [2]

Zou, H. (2006), "The Adaptive Lasso and its Oracle Properties," *Journal of the American Statistical Association*, 101, 1418–1429. [6]

Zuo, H., Lu, J., Zhang, G., and Liu, F. (2018), "Fuzzy Transfer Learning Using an Infinite Gaussian Mixture Model and Active Learning," *IEEE Transactions on Fuzzy Systems*, 27, 291–303. [2]